MIL-G-82445(OS)
10 May 1966
Superseding
WS4826
8 April 1966

MILITARY SPECIFICATION

GUIDANCE CONTROL TEST SET

MARK 443 MOD 1

(For Torpedo MK 46 MOD 0 and MOD 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers requirements for the procurement of the Guidance Control Test Set, Mark 443, MOD 1 herein referred to as the test set. (See 6.1 for description and intended use.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Fiber

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Military

MIL-P-116	Preservation, Methods of
MIL-S-901	Shock Tests, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-16400	Electronic Equipment, Naval Ship and Shore, General Specification
MIL-T-82426	Torpedo, Main Assembly

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-167	Mechanical Vibration of Shipboard Equipment

DRAWINGS

Bureau of Naval Weapons

DL 2540800	Guidance Control Test Set, and all documents listed thereon

PUBLICATIONS

Bureau of Naval Weapons

WD 28954	Special Manufacturing Procedure for Calibration of the Guidance Control Test Set
NAVWEPS OP 2656	Torpedo, Mark 46 Mod 0, Operation and Maintenance
NAVWEPS OP 3389	Torpedo, Mark 46 Mod 1, Description, Operation and Maintenance

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

PUBLICATIONS

Consolidated Classification Committee

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample shall:

(a) Be manufactured using procedures proposed for the production lot.

(b) Consist of two Guidance Control Test Sets.

(c) One test set to be tested as specified in section 4 herein and is for the purpose of determining that, prior to starting production, the supplier's production methods are capable of yielding items that comply with technical requirements of the contract.

(d) The second test set to be supplied to an activity designated by the procuring activity for the purpose of performance verification.

3.2 Basic Requirements.

3.2.1 Contractor-furnished data. No data is required by this specification, or by applicable documents referenced in section 2, unless specified in the contract or order (see 6.2).

3.2.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in manufacture of the Guidance Control Test Set shall be in accordance with requirements specified herein and with the applicable documents listed in BuWeps DL 2540800.

3.3 Performance requirements and product characteristics.

3.3.1 Input power requirements. The input power requirements shall be as specified in table I.

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Table I

Input Power

Voltage	Maximum Current
40 ±5 volts direct current (vdc)	2 amperes (amps) steady state, 45 amps pulsed for 40 milliseconds (ms).
-40 ±5 vdc	5 amps steady state, 45 amps pulsed for 40 ms.

Note: Necessary resolution of adjustment for the above input voltages shall be ±1 percent.

3.3.2 Input power switching and overvoltage protection.

3.3.2.1 Input power switching. The test set shall provide means for switching the input voltages to the test set and the guidance and control system being tested on and off. Indicator lamps (2) shall be included showing when these voltages are on.

3.3.2.2 Overvoltage protection. The test set shall provide internal circuitry to limit the dc input voltages to a preset threshold voltage from 42 to 44 vdc. When the input voltages exceed the trip level specified above, power to the test set shall be disabled.

3.3.3 Performance requirements. The test set shall provide the following test functions consisting of: input and output signal measurement; input and output voltage monitoring; simulated functions and loads required to evaluate the performance of a torpedo guidance and control system. The test set circuits shall switch automatically to the proper configuration for the type of guidance and control system (Mk 46 Mod 0 or Mk 46 Mod 1) being tested.

3.3.3.1 Time count switch. A time count switch makes it possible to measure 23 electrical functions with the aid of a bistable multivibrator. Output voltage of the multivibrator during reset shall be 0 to +2.0 vdc, and during count period +22.5 ±2.5 vdc.

3.3.3.2 Oscillator amplifier speed control. A variable frequency voltage from less than 1700 to greater than 2240 cps at a constant voltage of 32 ±4 volts root mean square (vrms) into a 3600 ohm load.

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3.3.3.3 Fire control voltages.

3.3.3.3.1 Search and ceiling depth. A regulated variable dc voltage of 18.5 ±1.5 to 31.5 ±1.5 volts into a 5 ohm load for actuating fire control circuits in the Guidance and Control System. The regulator shall be capable of withstanding a short circuit condition and to limit current to 8 amperes maximum for a sufficient period of time to blow a five (5) ampere fuse.

3.3.3.3.2 Search depth switch, search depth set switch, and lamp indicator. A search depth switch which provides 8 search depth positions and an indicator lamp for determining that the selected search depth has been obtained.

3.3.3.3.3 Ceiling mode, ceiling mode set switch, and lamp indicator. A ceiling mode switch which provides 6 ceiling mode positions and an indicator lamp for determining that the selected ceiling depth has been obtained.

3.3.3.4 Indicator light circuits. Indicator lamps for indicating roll (CW and CCW), pitch (UP and DOWN), yaw (RIGHT and LEFT), speed control (INCREASE and DECREASE), and COMPUTER ENABLE outputs of the guidance and control system. Indicator lamps to show whether the test set is in the Mk 46 Mod 0 or Mk 46 Mod 1 configuration.

3.3.3.5 Gyro relay. A double pole double throw relay to provide a ground for the course gyroscope when the meter switch is in position 1 and the meter function switch is in position 2.

3.3.3.6 Continuity indicator. A panel lamp indicator to assure that during testing all cables are properly connected to their respective receptacles.

3.3.3.7 Inverter circuit. A means for inverting a 2 kilocycle (kc) square wave input when the meter switch is in position 12 and the meter function switch is in position 3.

3.3.3.8 Scope switch. A 24 position switch for monitoring various functions and outputs of the guidance and control system at the scope jack on the front panel and providing appropriate synchronizing signals for these functions at the sync jack on the front panel.

3.3.3.9 Pendulum gyro simulator. A simulated pendulum signal continuously variable from 0 to 12.3 ±0.3 vrms. This signal must be provided with the output in-phase or 180 degrees out-of-phase with the 26.0 vrms 4.8 kc input voltage.

3.3.3.10 Rate gyro simulator. A simulated rate gyro signal continuously variable from 0 to 10.65 ±0.35 vrms. This signal must be provided with the output in-phase or 180 degrees out-of-phase with the 26.0 vrms 4.8 kc input voltage.

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3.3.3.11 Course gyro simulator. A simulated course gyro signal continuously variable from 2.83 ±0.30 vrms to 23.20 ±0.30 vrms. This signal must be in-phase with the 26.0 vrms 4.8 kc input voltage.

3.3.3.12 Gyro load. A 1 millihenry inductance paralleled with capacitance and resistance to obtain 290 ±15 ma at 22 ±4 degrees indicated phase shift with 26.0 vrms 4.8 kc input voltage.

3.3.3.13 Depth level simulator. A simulated depth voltage continuously variable from 0 to 23.11 ±0.46 vrms. This signal must be in-phase with the 26.0 vrms 4.8 kc input voltage.

3.3.3.14 Memory and test set calibration voltage. A +5.00 ±0.02 vdc and -5.00 ±0.05 vdc.

3.3.3.15 Transmitter control switching. Means for switching 5 inputs controlling 8 operating modes for the guidance and control system.

3.3.3.16 External ac voltage switching. Means for switching 23 ac signals to the EXT volts ac test jack on the front panel.

3.3.3.17 400 cps load. A 200 ohm load for the 26.0 vrms 400 cps power supply and a means for filtering the voltage waveform and reducing the output to 5.5 ±2.5 vrms for measuring frequency.

3.3.3.18 Exploder arm. A 35 ohm ±1 percent 80 watt load for the exploder arm circuit and an indicator lamp to light when the exploder arm voltage is supplied.

3.3.3.19 Voltmeter switching. Means for switching the panel meter to monitor 46 dc voltages generated either in the test set or the guidance and control system and to provide 23 special switching functions.

3.3.3.20 Alternator regulator. Provide circuitry to show that the alternator regulator will regulate the 40 volt torpedo power.

3.3.3.21 Yaw and pitch angle data. Yaw and pitch angle signal limiting for accurately measuring steering rates. Limiting voltage shall be: 1.2 ±0.1 vdc for either yaw data or pitch data input of 10 volts through 8200 ohms.

3.3.3.22 External transmitter load. The following resistive loads: 1.17 ohms ±2 percent 50 watts (2); 6.57 ohms ±2 percent 50 watts (2); 1.95 ohms ±2 percent 50 watts (2); 150 ohms ±5 percent 1/2 watt (4); and 1000 ohms ±5 percent 1/2 watt.

3.3.4 Life.

3.3.4.1 Service life. The service life of the test set shall be 5 years of normal service with a minimum of 60 days between calibration.

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3.4 Environmental requirements. Unless otherwise specified the test set shall be capable of conforming to the performance and product characteristics of this specification, after being exposed, at any temperature in the applicable range, to the environmental requirements under all conditions specified in this section.

3.4.1 Vibration. The test set shall be capable of withstanding the type I vibration required by Standard MIL-STD-167, except that the variable frequency requirement shall be as follows: The test set in operating condition shall be vibrated from 5 to 15 cps at a table amplitude of 0.030 ±0.006 inch and from 16 to 25 cps at a table amplitude of 0.020 ±0.004 inch. The vibration shall be applied in discrete intervals of 1 cps and applied for one minute at each interval.

3.4.2 Shock. The test set shall be capable of withstanding the shock required by Specification MIL-S-901, for Grade I, Type A, lightweight equipment, in the nonoperating condition, except that the height of the hammer drops shall be; 1 foot, 1-1/2 feet and 3 feet.

3.4.3 Inclination. The test set shall be capable of meeting the inclination test specified in Specification MIL-E-16400.

3.4.4 Temperature. The test set shall be capable of withstanding the temperatures specified in Specification MIL-E-16400 for Class 4 equipment.

3.4.5 Humidity. The test set shall be capable of withstanding the humidity requirements specified in Specification MIL-E-16400.

3.4.6 Accelerated life. The test set shall be capable of withstanding the accelerated life tests of Specification MIL-E-16400.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the test set shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. Inspections of the test set shall be classified as follows:

(a) Preproduction inspection

(1) Preproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

(c) Acceptance inspection

(1) Acceptance sampling (see 4.3.3)

(2) Acceptance tests and examinations (see 4.7)

4.3 Sampling.

4.3.1 Preproduction sample. A preproduction sample manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the test set by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sampling. Periodic production tests detailed in 4.6 shall be performed at an activity designated by the procuring activity. Samples shall consist of units that have satisfactorily passed the acceptance tests and examinations detailed in 4.7; selected as follows:

(a) Step 1. Upon initiation of production, select 1 unit at random from each group of 20 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to step 2.

(b) Step 2. Select 1 unit at random from each group of 50 units. Continue with step 2 for the balance of the contract or until a defective unit is found.

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4.3.2.1 Defective units. When a defective unit is found in any group the supplier shall:

(a) Determine the cause of the defects.

(b) Determine if the defect is an isolated case.

(c) Take necessary action to eliminate the defects from units on hand, and if so directed by the procuring activity, from those units previously shipped.

(d) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(e) Return to step 1 (4.3.2(a)) and proceed as indicated for inspection of units subsequently produced.

4.3.3 Acceptance sampling. The tests and examinations specified in 4.7 shall be conducted on each test set to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the test set shall be subjected to tests under the following conditions:

(a)	Temperature	Room ambient (60 to 95 degrees F)
(b)	Humidity (Relative)	Room ambient, up to 95 percent
(c)	Altitude	Normal ground atmospheric pressure
(d)	Vibration	None

4.5 Preproduction tests. The test set submitted as a preproduction sample shall be subjected to all preproduction tests specified in 4.5.2. Prior to and immediately after the preproduction tests the test set shall be subjected to the acceptance tests and examinations (see 4.7). Certified inspection data covering the results of preproduction tests shall be forwarded to the Bureau of Naval Weapons. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

(b) Run or test plan used.

(c) Serial number of the test set.

(d) Test results of preproduction tests.

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4.5.1 Preproduction test equipment. The following items of test equipment are required to perform the preproduction tests specified herein:

4.5.1.1 A vibration test fixture capable of applying vibration as specified in 3.4.1.

4.5.1.2 A shock fixture capable of applying shocks specified in 3.4.2.

4.5.1.3 An inclination testing device capable of performing the inclination tests specified in Specification MIL-E-16400.

4.5.1.4 A humidity-temperature chamber capable of applying the humidity and temperature tests specified in Specification MIL-E-16400.

4.5.1.5 Items of test equipment specified in 4.7.2.

4.5.2 Preproduction tests.

4.5.2.1 Vibrate the test set in accordance with vibration schedules required by Standard MIL-STD-167 with deviations specified in 3.4.1.

4.5.2.2 Shock. Secure the test set to the shock test fixture and apply shocks as specified in Specification MIL-S-901 with deviations specified in 3.4.2.

4.5.2.3 Inclination. Subject the test set to the inclination tests specified in Specification MIL-E-16400.

4.5.2.4 Temperature. Subject the test set to temperature tests, both operating and non-operating, as specified in Specification MIL-E-16400.

4.5.2.5 Humidity. Subject the test set to the humidity tests specified in Specification MIL-E-16400.

4.5.2.6 Accelerated life. Subject the test set to the accelerated life tests specified in Specification MIL-E-16400. Degradation of performance below specified requirements during this test, or inability to pass the acceptance test subsequent to this test, shall constitute a failure.

4.6 Periodic production tests. Certified inspection data covering the results of the periodic production tests shall be forwarded to the Bureau of Naval Weapons. The inspection data shall consist of items 4.5(a) through 4.5(c) and the results of the periodic production tests.

4.6.1 Periodic production test equipment. The following items of test equipment, or equivalent items, will be required to perform the tests specified herein:

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(a) Temperature chamber capable of providing the environment specified in 3.4.4.

(b) Vibration test fixture capable of applying vibration as specified in 3.4.1.

4.6.2 Periodic production tests. Each periodic production sample, after satisfactorily passing the acceptance tests and examination (4.7), shall be subjected to environmental tests for Temperature (4.5.2.4) and for Vibration (4.5.2.1). Unless otherwise specified, inability of the sample to pass satisfactorily the acceptance tests and examination subsequent to these environmental temperature and vibration tests, shall be deemed a defective unit.

4.7 Acceptance tests and examinations. Unless otherwise specified each Guidance Control test set delivered under contract shall be subjected to the tests and examinations specified herein.

4.7.1 Conditions.

4.7.1.1 Test sequence. Unless otherwise noted, sequence of testing procedures presented herein is mandatory and must be performed in order shown.

4.7.1.2 Test results. Where space is provided in Appendix Report Form to indicate value or specific reading obtained, the specific reading shall be recorded if a limit or limits is given in Limits column for the test. If limits are not stated in Limits column, the individual performing test or inspection shall initial in spaces provided in the Reading column indicating the requirements were satisfactorily met.

4.7.1.3 Acceptance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units which have not been corrected will be considered for acceptance only upon formal application by the contractor to the Bureau of Naval Weapons via the cognizant Government representative.

4.7.2 Acceptance test equipment. Items of test equipment listed in table II are required to perform acceptance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration. Record the verification of test equipment calibration.

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Table II

Acceptance Test Equipment

Item	Identification
Dual dc power supply (either output: 0 to 1, 10, 100 volts at 100 ma; line regulation: ±1% change in output for 95 to 125 vac input; ripple voltage: less than 0.01%)	ERA 110 MC
Oscilloscope (bandpass: dc to 15 mc; rise time: 0.023 micro-seconds; accuracy: 0.05 to 20 v/cm, 3%)	Tektronix, Model 535 with Type C-A Preamplifier
Dc vacuum tube voltmeter (range: 1 mv to 1,000 volt full scale; accuracy: ±1% full scale)	Hewlett-Packard Model 412A
Ac vacuum tube voltmeter (voltage range: 1.0 mv to 300v full scale; accuracy 2%; frequency range: 10 cps to 4 mc/s)	Hewlett-Packard Model 400D
Frequency counter (range: 10 cps to 120 kc/s; accuracy: ±1 count)	Hewlett-Packard Model 522B
Square wave generator (range: 1 cps to 1 mc; low impedance output; negative, 3.5 volts across 75 ohm impedance; high impedance output: negative, 27 volts across 600 ohm impedance; rise time: less than 0.1 usec)	Hewlett-Packard Model 211A
4.8 Kc power supply (frequency stability: less than ±0.25%; distortion: less than 1/2%; output power: 20 watts; voltage output: 0 to 120 volts at 800 ohm load; 0 to 60 volts at 200 ohm load; 0 to 28 volts at 40 ohm load)	Industrial Test Equip., Model 1500 (4.8 kc)

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Table II (Continued)

Acceptance Test Equipment

Item	Identification
Power oscillator (output voltage - tapped 0-120, 0-60, 0-28; output pwr - 400 cps ±0.15%; stability - after warmup, the stabilized freq. should not change more than 0.25% with line volt-variation ±10v; distortion - less than 1/2%; hum level - approx. 0.02% of rated output volt)	Industrial Test Equip., Model 1500 (400 cps)
Multimeter (resistance: 0 to 20 megohms; current: 0 to 10 amps; voltage: 0 to 5,000 ac or dc, accuracy: ±3% full scale for dc, ±5% full scale for ac)	Simpson Model 260
40 vdc power supply (2 required) (40 vdc at 15 amperes; range 0 to 50 vdc)	Trans Electronics Model No. RS-199
Ac dc digital voltmeter (display: 4 digits, polarity, decimal point, ac and A/B symbol; accuracy: 0.012%; bridge linearity: 0.003%, bridge resistance: 50,000 ohms; voltage range: 0 to 1000 volts)	Cubic Model V-41 Voltmeter Unit Cubic Model C-1 Control Unit Cubic Model AC-1 Converter Unit
Ac current probe (sensitivity: 1 mv/ma at 1 kc; response: ±2%, 100 cps to 3 mcs; rise time: less than 0.020 usec)	Hewlett-Packard Model 456A
Variable load resistor, (variable from 2 ohms to 6 ohms, 300 watts 10 amps (F/C power supply)	
Load resistor: 3600 ohms, 1/2 watt	
Load resistor: 680 ohms, 1/2 watt 1 percent	
Load resistor: 620 ohms, 1/2 watt 1 percent	

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Table II (Continued)

Acceptance Test Equipment

Item	Identification
Load resistor: 10,000 ohms, 1/4 watt, (5 required)	
Load resistor: 2 ohms non-inductive	
Pulse circuit for multivibrator (See figure 3.)	
Voltage divider resistors: 2400, 1500, 2000 and 1000 ohms, all 1/2 watt; 2000 ohm potentiometer, 1 watt (See figure 4.)	
Load resistor: 10.0 ohms, ±1%, 200 watts	
Load resistor: 20.0 ohms, ±1%, 100 watts	
Load resistor: 8200 ohms, 1/2 watt	
Impedance bridge (Ranges: Resistance: 0.001 ohm to 10 megohms; capacitance: 1 micro-microfarad to 1,000 microfarads; inductance: 1 microhenry to 1000 henrys. Accuracy: resistance ±1%, capacitance ±1%, inductance ±1%; frequency range: 1% accuracy for L and C, 20 cps to 20 kc/s, for R, 10 cps to 50 kc/s)	General Radio Model 1650-A
Capacitor: 0.1 ufd, 400 vdc	
Dc ammeter: 0 to 5 amperes ±5%	
Phase meter (frequency range: 8 cps to 100 kc/s; phase range: 0 to 360 degrees in 8 ranges; input signal amplitude: 0.3 to 70.0 vac; accuracy: 0.25 degree relative, ±1 degree of ±2 percent absolute)	Ad-Yu Model 405

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4.7.3 Electrical tests.

4.7.3.1 Test setup.

4.7.3.1.1 Guide pin orientation. Check guide pin orientation of connectors J17, P17, J2350; J14, P14, J2351; J18, P18, P2302A, P2302B, J28, P28, P5511; J15, P15, P2323, P3602, J4202 and P2324 shown in figure 1.

4.7.3.1.2 Continuity. Connect cables to distribution box as shown in figure 1. Connect the ohmmeter between the listed connector pins. The resistance value shall not exceed 5 ohms for each measurement.

Connector	Pin		Connector	Pin
P2323 -	A	to	P4 -	W
	B			X
	C			Y
	D			Z
	E			a
	F			b
	H			c
	J			d
	K			e
	L			f
	M			g
	N			h
	P			i
	R			j
	T			m
	U			n
P2323 -	V	to	P4 -	p
P2324 -	C	to	P4 -	HH
	E			q
	F			r
	H			GG
P2324 -	J	to	P4 -	s

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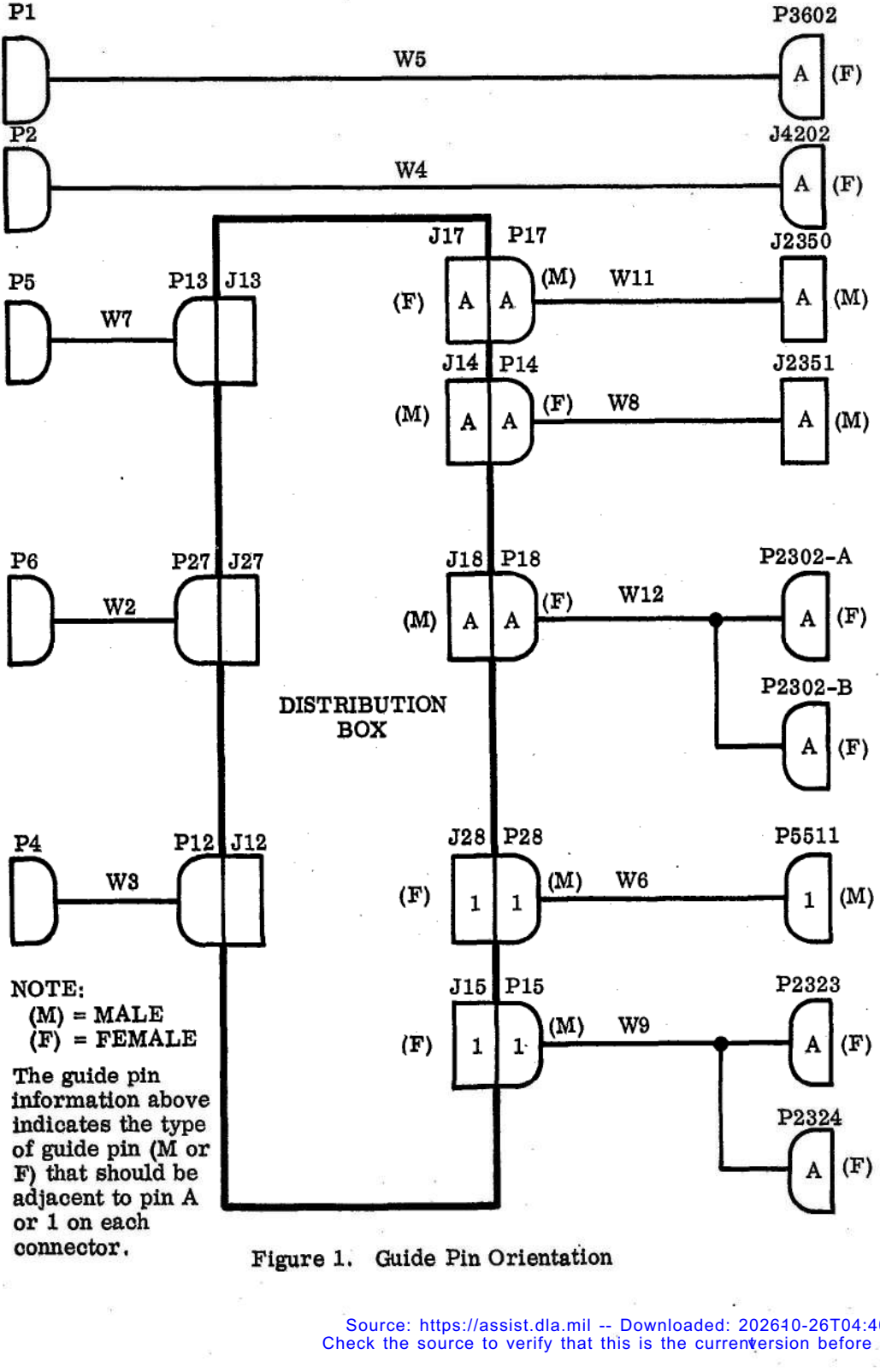


Figure 1. Guide Pin Orientation

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4.7.3.1.2 (Cont)

Connector	Pin		Connector	Pin
P2324 -	N	to	P4 -	t
	P			u
	R			v
	S			w
	T			x
	U			y
	V			EE
P2324 -	X	to	P4 -	z

P2302A-	A	to	P4 -	A
	C			C
	D			D
	E			E
	H			G
	J			H
	K			J
	P			M
	R			N
	S			P
	T			R
P2302A-	V	to	P4 -	T

J2351 -	A	to	P2350 -	A
	B		P2350 -	B
	B		P5 -	u
	C		P5 -	t
	E		P5 -	s
	F		P2350 -	F
	F		P5 -	r
	J		P2350 -	J
J2351 -	J	to	P5 -	q

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4.7.3.1.2 (Cont)

Connector	Pin		Connector	Pin
J2351 -	K	to	P5 -	y
	L		P2350 -	L
	L		P5 -	n
	M		P2350 -	M
	N		P2350 -	N
	P		P2350 -	P
	P		P5 -	m
	R		P2350 -	R
	S		P2350 -	S
	S		P5 -	k
	T		P2350 -	T
	T		P5 -	i
	U		P2350 -	U
	U		P5 -	h
	V		P2350 -	V
	W		P2350 -	W
	W		P5 -	g
	X		P2350 -	X
	X		P5 -	f
	Z		P2350 -	Z
	Z		P5 -	e
	b		P5 -	j
	b		P5 -	d
	d		P5 -	p
	f		P2350 -	f
	g		P5 -	L
	h		P2350 -	h
	h		P5 -	c
	i		P2350 -	i
	j		P2350 -	j
J2351 -	k	to	P2350 -	k

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4.7.3.1.2 (Cont)

Connector	Pin		Connector	Pin
J2351 -	m	to	P2350 -	m
	n		P2350 -	n
	n		P5 -	b
	p		P5 -	a
	q		P2350 -	q
	q		P5 -	Z
	s		P2350 -	s
	s		P5 -	Y
	t		P2350 -	t
	t		P5 -	X
	u		P2350 -	u
	v		P2350 -	v
	x		P5 -	w
	y		P2350 -	y
	z		P2350 -	z
	z		P2302B -	S
	AA		P2350 -	AA
	AA		P5 -	W
	AB		P2350 -	AB
	AC		P2350 -	AC
	AE		P2350 -	AE
	AE		P5 -	V
	AF		P2350 -	AF
	AF		P5 -	U
	AH		P2350 -	AH
	AH		P5 -	T
	AJ		P2350 -	AJ
	AN		P2350 -	AN
	AN		P5 -	S
	AP		P2350 -	AP
J2351 -	AP	to	P5 -	P

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4.7.3.1.2 (Cont)

Connector	Pin		Connector	Pin
J2350 -	K	to	P5 -	x
	x			v
	d			H
J2350 -	p	to	P5 -	G
P5511 -	1	to	P6 -	A
	3			C
	5			D
	9			F
	10			G
	11			H
	15			K
	29			S
	30			L
	34			T
	35			U
	36			W
	40			X
	41			Y
	45			a
	47			d
	50			f
	51			g
	52			h
	56			k
	57			m
	58			n
	59			p
	60			q
	62			r
	63			s
P5511 -	65	to	P6 -	v

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4.7.3.1.2 (Cont)

Connector	Pin		Connector	Pin
P5511 -	Terminal Lug	to	P6 -	V
	28			P
	22			O
	38			N
P5511 -	23	to	P6 -	Q
J4202 -	L	to	P2 -	A
	U			B
	T			C
	S			D
	K			E
	P			F
	D			G
	J			H
	A			J
	C			K
	B			L
	R			M
	M			N
	F			P
	E			R
J4202 -	W	to	P2 -	S
P3602 -	A	to	P1 -	A
	B			B
	C			C
	D			D
	E			E
	J			J
	H			H
	P			P
P3602 -	L	to	P1 -	L

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4.7.3.1.3 Power supply outputs. Connect the front panel output terminals of the +40 volt and the -40 volt power supplies to the vtvm and monitor output. Output of each supply shall be 39 to 41 volts dc. If necessary, adjust the vernier controls to attain the required output voltage.

4.7.3.2 Performance tests (simulated).

4.7.3.2.1 Over voltage protection.

CAUTION

During the following tests do not increase the supply voltage above 45.5.

4.7.3.2.1.1 On the test set front panel, place all power switches in the OFF position. TEST SET POWER, TORPEDO POWER and FIRE CONT POWER. Connect the +40 volt power supply between J3 pin B(+) and J3 pin A(-). Connect the -40 volt power supply between J3 pin C(-) and J3 pin A(+). See figure 2.

4.7.3.2.1.2 Turn the TEST SET POWER switch to ON. The TEST SET POWER lamp shall illuminate.

4.7.3.2.1.3 Increase the output from the +40 volt supply (but not above +45.5 vdc) until the TEST SET POWER lamp extinguishes. The output voltage shall be +42.0 to +44.0 vdc.

4.7.3.2.1.4 Set the ALT REG LOAD switch to the IN position (switch is momentarily up). The TEST SET POWER lamp shall illuminate. Release switch and set the +40 volt supply to 40.0 ±0.4 vdc.

4.7.3.2.1.5 Increase the output from the -40 volt supply (but not above -45.5 vdc) until the TEST SET POWER lamp extinguishes. The output voltage shall be -42.0 to -44.0 vdc.

4.7.3.2.1.6 Set the ALT REG LOAD switch to the IN position (switch is momentarily up). The TEST SET POWER lamp shall illuminate. Release switch and set the -40 volt supply to 40.0 ±0.4 vdc. Connect the dc voltmeter to the TIME COUNT jack on the test set front panel.

4.7.3.2.1.7 On the front panel of the test set, set the controls or switches as shown below.

Function	Position
TEST SET, POWER	OFF
TORPEDO, POWER	OFF
FIRE CONT POWER	OFF

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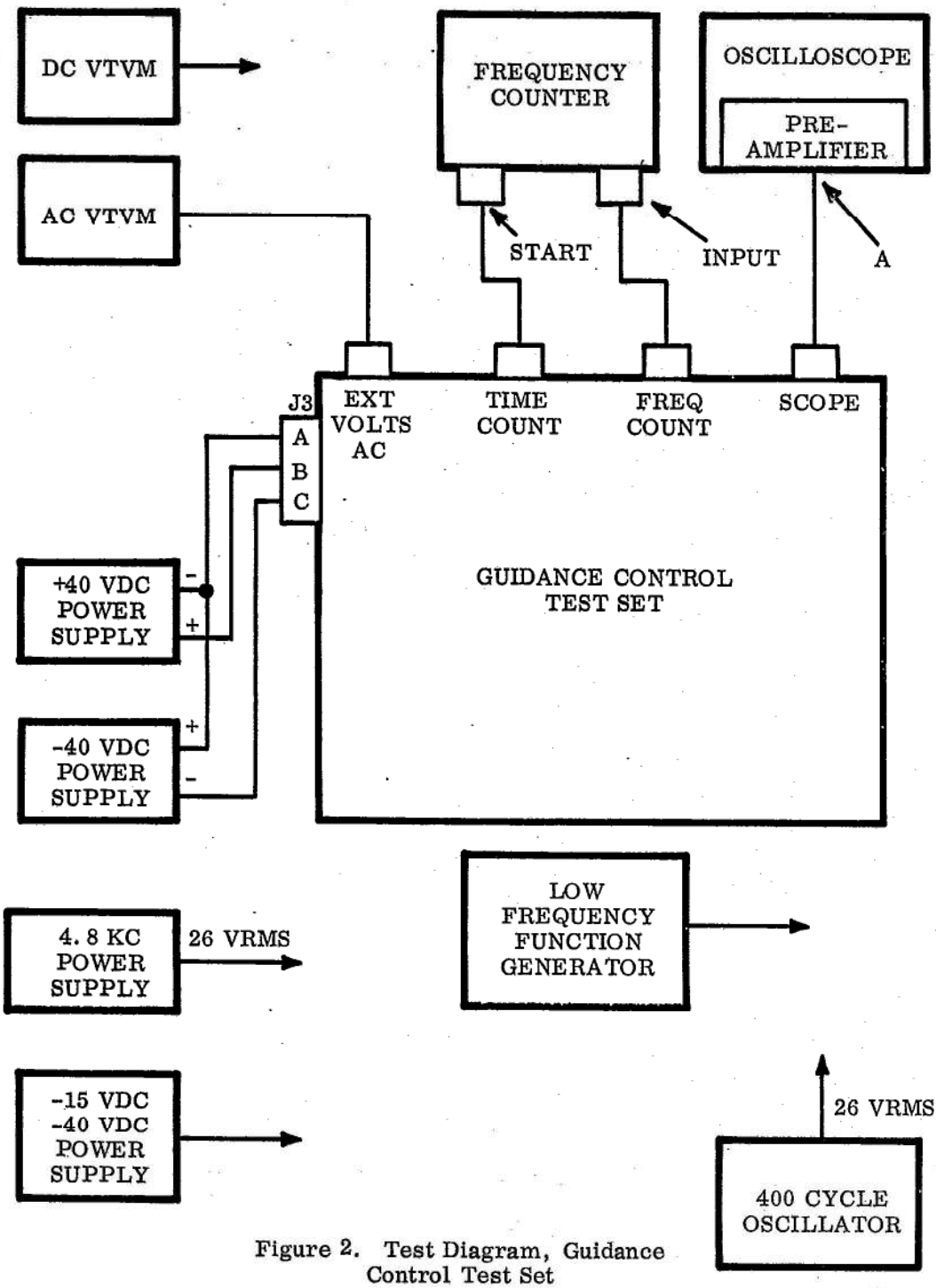


Figure 2. Test Diagram, Guidance Control Test Set

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4.7.3.2.1.7 (Cont)

Function	Position
METER SWITCH	OFF
METER FUNCTION	2
TUBE-AIR/ASROC	AIR/ASROC
SCOPE SWITCH	24
SEARCH DEPTH	1
CEILING MODE	1
EXT VOLTS AC	OFF
COUNT SWITCH	1
GYRO SELECTOR	ROLL
XMTR CONTROL SWITCH	6
DEPTH FUNCTION	OFF
FIRE CONTROL VOLTAGE	Center
COURSE GYRO	0
RATE	0
PENDULUM	0
ALT FREQ	Center
DEPTH, LEVEL X100	0

4.7.3.2.2 Bistable multivibrator circuit.

4.7.3.2.2.1 Connect jumpers from J6 pin j to J4 pin w and from J6 pin b to J4 pin v. Set the TEST SET POWER switch to ON. Place the TORPEDO POWER switch to ON.

4.7.3.2.2.2 Perform all tests in table III using the test circuit (pulse network) shown in figure 3. Connect the START and STOP outputs to the test set connectors as specified in column three of table III. Position the COUNT SWITCH to the various positions as shown in column two. In the pulse network, set switches S1 or S2 in the correct position to obtain proper polarity of the pulse as listed in column four (see figure 3). Monitor J7 (TIME COUNT) with a dc voltmeter for the voltages listed under column five.

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NOTE

The COMPUTER ENABLE indicator lamp will be illuminated at switch positions 6 and 9

Table III

Bistable Multivibrator Circuit

Step No.	Test Set COUNT SW Position	Test Circuit Output Connections to Test Set	Test Circuit Switches P = Press R = Release	Voltage Indication (vdc)
1	1	START to J5-y STOP to J5-w	S1 to (-) S2 to (-) S3, P & R S4, P & R	 20 to 25 0 to 2
2	2	START to J5-w STOP to J5-h	 S3, P & R S4, P & R	 20 to 25 0 to 2
3	3	START to J5-q	S1 to (+) S3, P S3, R	 20 to 25 0 to 2
4	4	START to J5-h STOP to J4-n	 S1 to (-) S3, P & R S4, P & R	 20 to 25 0 to 2
5	5	STOP to J5-X	TORPEDO POWER to OFF, then ON S4, P & R	 20 to 25 0 to 2
6	6	STOP to J4-r	TORPEDO POWER to OFF, then ON S4, P & R	 20 to 25 0 to 2
7	8	STOP to J6-M	S1 to (+) S3, P & R S4, P & R	 20 to 25 0 to 2

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Table III (Continued

Bistable Multivibrator Circuit

Step No.	Test Set COUNT SW Position	Test Circuit Output Connections to Test Set	Test Circuit Switches P = Press R = Release	Voltage Indication (vdc)
8	9	START to J4-r STOP to J6-M	S1 to (-) S2 to (+) S3, P & R S4, P & R	 20 to 25 0 to 2
9	10	START to J5-h	 S3, P S3, R	 20 to 25 0 to 2
10	11	START to J6-M STOP to J5-e	S1 to (+) S3, P & R S4, P & R	 20 to 25 0 to 2
11	12	START to J5-X	S1 to (-) S3, P & R S4, P & R	 20 to 25 0 to 2
12	13	START to J6-J STOP to J4-r	S1 to (+) S2 to (-) S3, P & R S4, P & R	 20 to 25 0 to 2
13	14	START to J5-h STOP to J5-q	S1 to (-) S3, P & R S4, P & R	 20 to 25 0 to 2
14	16	START to J5-y	 S3, P S3, R	 20 to 25 0 to 2
15	17	START to J5-w	 S3, P S3, R	 20 to 25 0 to 2
16	18	START to J5-a	 S3, P S3, R	 20 to 25 0 to 2
17	19	START to J5-y STOP to J4-E	S1 to (+) S3, P & R S4, P & R	 20 to 25 0 to 2

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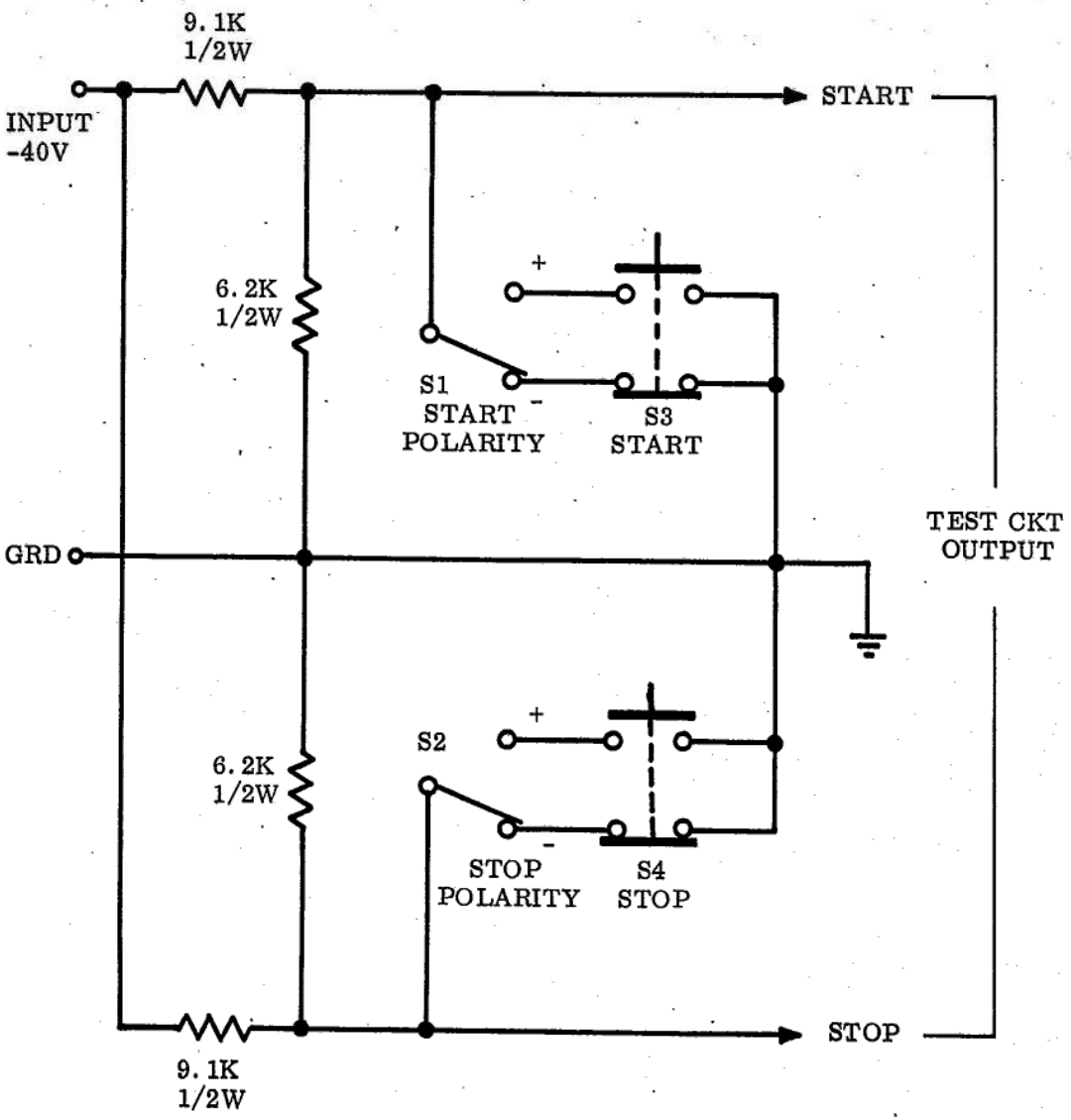


Figure 3. Pulse Network

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4.7.3.2.3 Bistable reset.

4.7.3.2.3.1 Connect the start lead on the test circuit (figure 3) to pin y (J5). Set the START POLARITY switch on the test circuit to (+).

4.7.3.2.3.2 On the test circuit press the START pushbutton. The dc vtvm shall indicate 20 to 25 volts dc.

4.7.3.2.3.3 Turn the METER FUNCTION switch to position 3. Dc vtvm indication shall be -0.5 to +0.5 (0.0 ±0.5) volt dc. Return METER FUNCTION switch to position 2.

4.7.3.2.4 Diode steering.

4.7.3.2.4.1 Disconnect the power supply and test circuit used for the bistable multivibrator circuit test. Do not disconnect the dc vtvm.

4.7.3.2.4.2 Set COUNT SWITCH to position 15. Set the output of the function generator to obtain a square wave output with a repetition frequency of 1 cps at 15 volts peak-to-peak ±1 volt.

4.7.3.2.4.3 Connect the square wave output to pin q of J5 and pin A (ground) of J6. Monitor the dc vtvm. The vtvm voltage indication shall change from approximately 1 volt to 25 volts at one half the repetition rate of the function generator (1 cps).

4.7.3.2.4.4 Disconnect the lead to pin q of J5 and connect the lead to pin w of J5. Set COUNT SWITCH to position 7. The vtvm shall indicate a change from approximately 1 volt to 25 volts at one half the repetition rate of the function generator (1 cps).

4.7.3.2.4.5 Disconnect the leads from the function generator and dc vtvm. Turn the power to the function generator to OFF. Leave the TEST SET POWER and the TORPEDO POWER switches ON.

4.7.3.2.5 Oscillator amplifier speed control circuit.

4.7.3.2.5.1 On the test set front panel, set the COUNT SWITCH selector to ALT FREQ position. Connect a 3600 ohm 1/2 watt load resistor across pins O and P of connector J6.

4.7.3.2.5.2 Connect the electronic counter to FREQ COUNT jack J8 and measure the frequency with the ALT FREQ control rotated to the maximum INCREASE position. The counter indication shall be greater than 2240 cps.

4.7.3.2.5.3 Rotate the ALT FREQ control to the maximum DECREASE position. The counter indication shall be 1200 to 1700 cps.

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4.7.3.2.5.4 Position the EXT VOLTS AC selector switch to position 1 and connect the ac vtvm to J11 EXT VOLTS AC. The voltage indication shall be 28 to 36 (32 ±4) volts rms.

4.7.3.2.5.5 Disconnect the load resistor, vtvm and counter. Turn the TEST SET POWER switch to OFF.

4.7.3.2.6 Fire control voltage regulator circuit.

4.7.3.2.6.1 Adjust variable load resistor (2 to 6 ohms) for 6 ohms. Connect one side of the load to J6 pin k. Connect multimeter to measure current on 10 ampere scale. Connect the positive lead of the multimeter to J6 pin f and the negative lead to the other side of the variable load.

4.7.3.2.6.2 Set the METER SWITCH to position 15. Set the METER FUNCTION switch to position 2.

4.7.3.2.6.3 Turn the TEST SET, TORPEDO and FIRE CONT POWER switches ON.

NOTE

In steps 4.7.3.2.6.4 and 4.7.3.2.6.5 the CEILING MODE and SEARCH DEPTH lamps will be illuminated. The TORPEDO POWER lamp will be extinguished.

4.7.3.2.6.4 With the FIRE CONTROL VOLTAGE control at the full INCREASE position, the voltage displayed on the panel meter shall indicate 30 to 33 volts dc.

4.7.3.2.6.5 With the FIRE CONTROL VOLTAGE control at the full DECREASE position, the panel meter shall indicate 17 to 20 volts dc.

4.7.3.2.6.6 Turn the TEST SET POWER switch OFF. Change load resistance from 6 ohms (paragraph 4.7.3.2.6.1) to 3.5 ohms.

CAUTION

The fire control fuse holder requires a 5 ampere fuse. Do not allow current limiting condition to prevail for more than 2 seconds. If fuse blows replace with 5 ampere fuse.

4.7.3.2.6.7 Turn TEST SET POWER switch ON. Observe current indication on multimeter. Increase the FIRE CONTROL VOLTAGE control steadily until current stops increasing abruptly. The current indication at that point shall be 6 to 8 amperes.

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4.7.3.2.6.8 Turn TEST SET POWER switch OFF. Disconnect the variable load and the multimeter.

4.7.3.2.7 Fire control (search depth).

4.7.3.2.7.1 Set the METER SWITCH to position 17, the METER FUNCTION switch to position 2. Set the TUBE-AIR/ASROC switch to TUBE position. Connect a 680 ohm 1/2 watt resistor between J6 pin C and J6 pin a.

4.7.3.2.7.2 Set the FIRE CONTROL VOLTAGE potentiometer to FULL INCREASE position, the SEARCH DEPTH switch to position 1. Turn the TEST SET POWER switch ON.

NOTE

In steps 4.7.3.2.7.3 through 4.7.3.2.7.8 the SEARCH DEPTH lamp may display a momentary flash at the time fire control power is applied. CEILING MODE lamp will be illuminated. The TORPEDO POWER lamp will extinguish.

4.7.3.2.7.3 Turn the FIRE CONTROL POWER switch ON. The SEARCH DEPTH lamp shall be extinguished.

4.7.3.2.7.4 Connect a dc vtvm between J6 pins k and F. Depress the SEARCH DEPTH SET switch. The vtvm shall indicate 30 to 33 volts dc.

4.7.3.2.7.5 Set the SEARCH DEPTH switch to position 2. The SEARCH DEPTH lamp shall remain extinguished.

4.7.3.2.7.6 Set the SEARCH DEPTH switch to position 3. The SEARCH DEPTH lamp shall remain extinguished.

4.7.3.2.7.7 Set the SEARCH DEPTH switch to position 4. The SEARCH DEPTH lamp shall remain extinguished.

4.7.3.2.7.8 Set the SEARCH DEPTH switch to position 5. The SEARCH DEPTH lamp shall remain extinguished.

4.7.3.2.7.9 Set the SEARCH DEPTH switch to position 6. The SEARCH DEPTH lamp shall remain extinguished.

4.7.3.2.7.10 Set the SEARCH DEPTH switch to position 7. The SEARCH DEPTH lamp shall illuminate.

4.7.3.2.7.11 Set the SEARCH DEPTH switch to TEST. The SEARCH DEPTH lamp shall remain illuminated.

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4.7.3.2.7.12 With the FIRE CONT POWER switch in the ON position and the SEARCH DEPTH switch in the TEST position, the front panel meter shall display the fire control voltage which may be varied with the FIRE CONTROL VOLTAGE adjustment from approximately 20 to 30 volts dc.

4.7.3.2.7.13 Turn the TEST SET POWER switch OFF and disconnect the 680 ohm load resistor.

4.7.3.2.8 Fire control (ceiling mode).

4.7.3.2.8.1 Set the METER SWITCH to position 16, the TUBE-AIR/ASROC switch to the AIR/ASROC position. Connect a 620 ohm, 1/2 watt resistor between J6 pin f and J6 pin r. Set the FIRE CONTROL VOLTAGE control to FULL CW or INCREASE position. Set the CEILING MODE switch to position 1.

NOTE

In steps 4.7.3.2.8.3 through 4.7.3.2.8.8 the CEILING MODE lamp may display a momentary flash at the time the fire control power is applied. The SEARCH DEPTH lamp shall be illuminated. The TORPEDO POWER lamp shall be extinguished.

4.7.3.2.8.2 Turn the TEST SET POWER and FIRE CONT POWER switches ON. The CEILING MODE lamp shall be extinguished.

4.7.3.2.8.3 Connect the vtvm between J6 pins k and K. Depress CEILING MODE SET switch. The vtvm shall indicate 30 to 33 volts dc.

4.7.3.2.8.4 Set the CEILING MODE switch to position 2. The CEILING MODE lamp shall remain extinguished.

4.7.3.2.8.5 Set the CEILING MODE switch to position 3. The CEILING MODE lamp shall remain extinguished.

4.7.3.2.8.6 Set the CEILING MODE switch to position 4. The CEILING MODE lamp shall remain extinguished.

4.7.3.2.8.7 Set the CEILING MODE switch to position 5. The CEILING MODE lamp shall illuminate.

4.7.3.2.8.8 Set the CEILING MODE switch to TEST. The CEILING MODE lamp shall remain illuminated. The front panel meter shall indicate 25 to 35 volts.

4.7.3.2.8.9 Disconnect the 620 ohm load resistor. Turn the FIRE CONTROL POWER switch OFF.

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4.7.3.2.9 Light circuits.

4.7.3.2.9.1 Connect a jumper between J6 pin j and J4 pin w (-40v); also between J6 pin b and J4 pin v (+40v). Leave the jumpers connected during light circuit tests.

4.7.3.2.9.2 Position the TEST SET POWER and TORPEDO POWER switches ON. Note that the STEERING COMMAND indicator lamps RIGHT, CW and DOWN are illuminated.

4.7.3.2.9.3 Position the METER SWITCH to 9, the FUNCTION switch to 2. The indication on the panel meter shall be 37 to 43 volts dc.

4.7.3.2.9.4 Connect a jumper from J6 pins G to J. The DOWN lamp shall be extinguished. The UP lamp shall be illuminated. The panel meter shall indicate -0.5 to +0.5 (0.0 ±0.5) volt dc.

4.7.3.2.9.5 Disconnect the jumper between pins G and J on J6. Position the METER SWITCH to 10. The panel meter shall indicate 37 to 43 volts dc.

4.7.3.2.9.6 Connect a jumper between J6 pins Q and E. The STEERING COMMAND CW ROLL lamp shall extinguish. The STEERING COMMAND CCW ROLL lamp shall illuminate. The panel meter shall indicate -0.5 to +0.5 (0.0 ±0.5) volt dc.

4.7.3.2.9.7 Disconnect the jumper between pins Q and E on J6. Position METER SWITCH to 11. The panel meter shall indicate 37 to 43 volts dc.

4.7.3.2.9.8 Connect a jumper between J6 pins A to M. The STEERING COMMAND RIGHT lamp shall extinguish. The STEERING COMMAND LEFT lamp shall illuminate. The panel meter shall indicate -0.5 to +0.5 (0.0 ±0.5) volt.

4.7.3.2.9.9 Turn the TEST SET POWER switch OFF. Disconnect the jumper between pins A and M on J6.

4.7.3.2.9.10 Using the test circuit (voltage divider) shown in figure 4, connect lead A to J6 pin e and lead B to J6 pin Z. Leave the jumpers connected as in 4.7.3.2.9.1. Turn the TEST SET POWER and TORPEDO POWER switches ON.

4.7.3.2.9.11 Adjust R1 on the test circuit to indicate zero volts on the dc vtvm. Turn R1 slowly CW until the dc vtvm indicates approximately +1 volt. The SPEED CONTROL INCREASE lamp shall illuminate.

4.7.3.2.9.12 Turn R1 to full CW position. The SPEED CONTROL INCREASE lamp shall remain illuminated.

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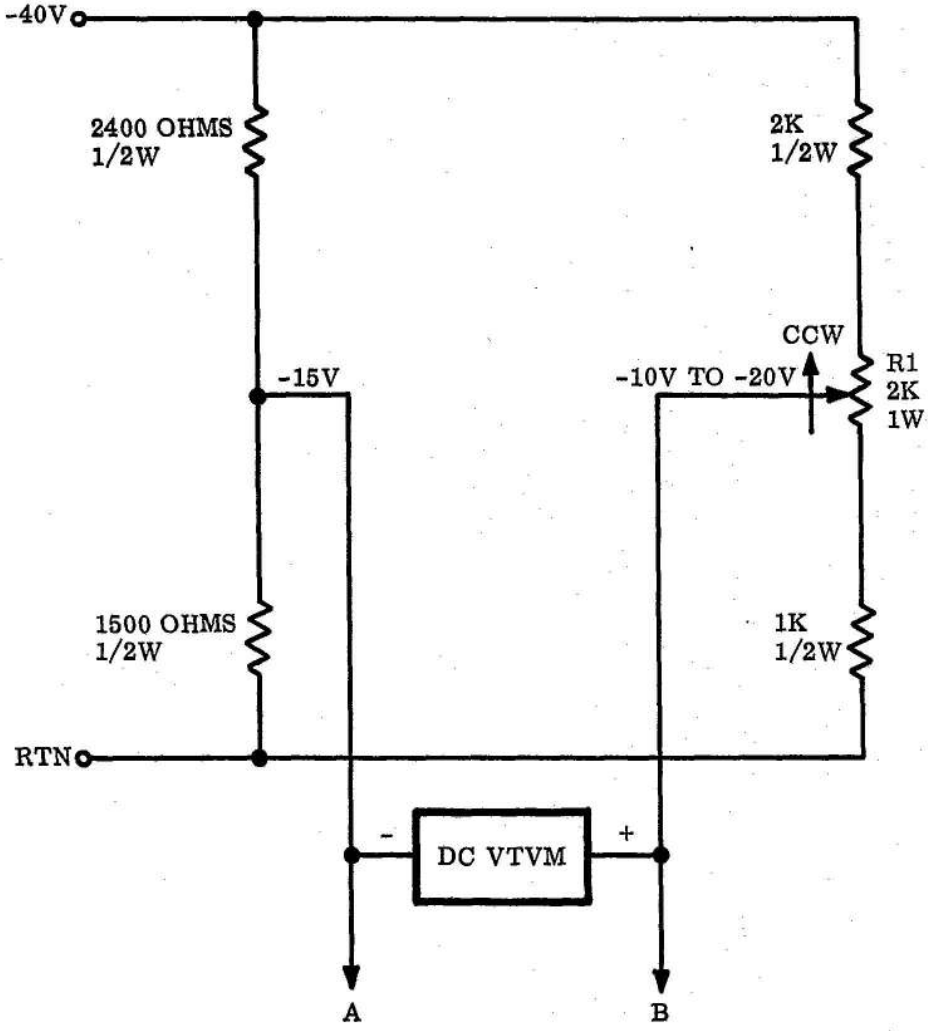


Figure 4. Voltage Divider Circuit

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4.7.3.2.9.13 Set R1 to indicate zero volts on the dc vtvm. Reverse the leads on the dc vtvm to indicate negative voltage. Set R1 to indicate zero volts on the dc vtvm. Turn R1 CCW slowly until the dc vtvm indicates approximately -1 volt. The SPEED CONTROL DECREASE lamp shall illuminate.

4.7.3.2.9.14 Turn R1 to full CCW position. The SPEED CONTROL DECREASE lamp shall remain illuminated.

4.7.3.2.9.15 Turn the TEST SET and TORPEDO POWER switches to OFF. Disconnect the test circuit and all jumper wires. Disconnect all cables except W2 from the Distribution Box. Connect cable W2 from J6 on the test set to J27 on the distribution box. Turn the TEST SET POWER switch ON. The distribution box MOD 0 lamp shall illuminate. Depress MOD 0 "Push to test" lamp; it shall illuminate. Depress MOD 1 "Push to test" lamp; it shall illuminate.

4.7.3.2.9.16 Turn TORPEDO POWER switch ON. Connect dc vtvm from J28 pin 23 (ground) to the pins listed. Indications shall be as follows:

J28 pin	Indication (vdc)
53	-38.5 to -41.5
24	-38.5 to -41.5
46	+38.5 to +41.5

4.7.3.2.9.17 Turn TORPEDO and TEST SET POWER switches to OFF. Connect a jumper from J28 pin 46 to J4 pin v (+40 vdc), and J28 pin 53 to J4 pin w (-40 vdc). Turn TEST SET and TORPEDO POWER switches ON. Steering lamps RIGHT, CW, and DOWN shall illuminate.

4.7.3.2.9.18 Connect a jumper from J28 pins 10 and 12. The DOWN lamp shall extinguish, the UP lamp shall illuminate.

4.7.3.2.9.19 Disconnect jumper from J28 pin 12 and connect to J28 pin 6. The CW lamp shall extinguish, the CCW lamp shall illuminate.

4.7.3.2.9.20 Disconnect jumper lead from J28 pin 6 and connect to J28 pin 18. The RIGHT lamp shall extinguish, the LEFT lamp shall illuminate. Disconnect jumper wire from J28 pins 10 and 18.

4.7.3.2.9.21 Connect jumper to J28 pins 23 and 39. The MOD 0 lamp shall extinguish, the MOD 1 lamp shall illuminate.

4.7.3.2.9.22 Measure voltage between J28 pin 23 and J28 pins 54, 48, and 46. In each case the indication shall be +38.5 to +41.5 volts dc.

4.7.3.2.9.23 Measure voltage between J28 pin 23 and J28 pins 24, 49, 53 and 55. In each case the indication shall be -38.5 to -41.5 volts dc. Remove vtvm.

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4.7.3.2.9.24 Measure continuity between J14 pin D and J12 pin n. Continuity shall be 5.0 ohms maximum.

4.7.3.2.9.25 Connect a jumper to J28 pins 10 and 18; a second jumper to J28 pins 53 and 16. The LEFT lamp shall extinguish, the RIGHT lamp shall illuminate.

4.7.3.2.9.26 Disconnect jumper lead from J28 pin 53. Connect jumper to J28 pins 16 and 46. The RIGHT lamp shall extinguish, the LEFT lamp shall illuminate.

4.7.3.2.9.27 Disconnect jumper from J28 pin 10 to J28 pin 18 and from J28 pin 46 to J28 pin 16. Connect jumper to J28 pins 10 and 42 and a second jumper to J28 pins 46 and 43. The UP lamp shall extinguish, the DOWN lamp shall illuminate.

4.7.3.2.9.28 Disconnect jumper lead from J28 pin 46. Connect jumper to J28 pins 43 and 53. The DOWN lamp shall extinguish, the UP lamp shall illuminate. Disconnect jumpers from J28 pins 53 and 43; J28 pins 10 and 42. Turn the TORPEDO and TEST SET POWER switches OFF.

4.7.3.2.9.29 Set function generator for 400 cps square wave at 22 ±2.0 volts peak-to-peak and connect to test circuit as shown in figure 5. Turn TEST SET and TORPEDO POWER switches ON. Set switches in figure 5 as follows and note corresponding results. With TEST SET POWER switch in position 1 the CCW lamp shall extinguish and the CW lamp shall illuminate. With TEST SET POWER switch in position 2 the CW lamp shall extinguish and the CCW lamp shall illuminate. With TEST SET POWER switch in position 3 the CW lamp shall extinguish and the CCW lamp shall illuminate. With TEST SET POWER switch in position 4 the CCW lamp shall extinguish and the CW lamp shall illuminate.

NOTE

During roll test LEFT and RIGHT indicator lamps illuminate and extinguish.

4.7.3.2.9.30 Turn the TORPEDO and TEST SET POWER switch OFF. Remove test circuit of figure 5 and disconnect cable W2 from J6.

4.7.3.2.10 Gyro relay circuit.

4.7.3.2.10.1 Set the TEST SET POWER switch to ON and the METER SWITCH to OFF.

4.7.3.2.10.2 Connect the ohmmeter between J4 pin m and J6 pin Q. The indication shall be greater than 200,000 ohms.

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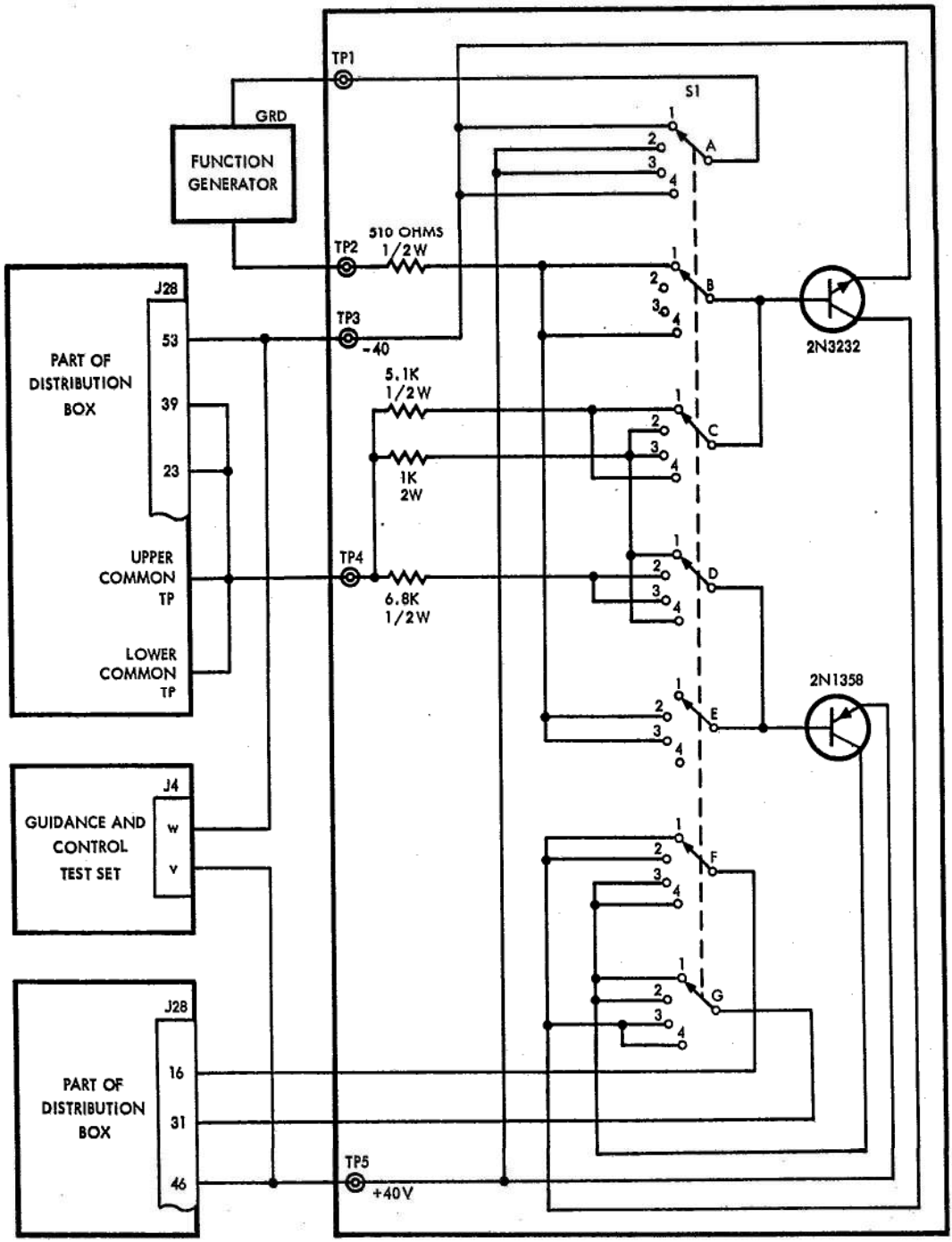


Figure 5. 400 Cycle Signal Circuit

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4.7.3.2.10.3 Turn the METER SWITCH to position 1. The ohmmeter indication shall be less than 5 ohms. Disconnect the ohmmeter.

4.7.3.2.11 Circuit continuity.

4.7.3.2.11.1 The SYSTEM CONTINUITY lamp (DS3) shall be extinguished at the start of test.

4.7.3.2.11.2 On front panel jack J6, connect a jumper between pins s and g. The SYSTEM CONTINUITY lamp shall illuminate.

4.7.3.2.11.3 Remove the jumper. Turn the TEST SET POWER switch to OFF. Connect an ohmmeter between pins R and S on J2. The indication shall be less than 5 ohms.

4.7.3.2.11.4 Connect the ohmmeter between pins J and H of J1. The indication shall be less than 5 ohms.

4.7.3.2.12 Inverter circuit.

4.7.3.2.12.1 Connect the output of the square wave generator (set at 12 volts peak-to-peak and 2 kc) through a 0.1 microfarad 400 volt capacitor to jack J4 pin x, and the generator signal return to plug J6 pin Q.

4.7.3.2.12.2 Turn the SCOPE SWITCH to position 13, the METER SWITCH to position 11, and the METER FUNCTION switch to position 3.

4.7.3.2.12.3 Connect the oscilloscope channel B to front panel SCOPE jack (J9). Connect the oscilloscope sync to SCOPE SYNC jack J10. Connect the oscilloscope channel A to the output of the square wave generator. Turn the TEST SET POWER switch to ON. The two oscilloscope traces shall be in phase.

4.7.3.2.12.4 Turn the METER SWITCH to position 12. The oscilloscope traces shall be approximately 180 degrees out of phase.

4.7.3.2.12.5 Disconnect the test equipment and jumpers.

4.7.3.2.13 Scope switch continuity.

4.7.3.2.13.1 Turn the TEST SET POWER switch to OFF. On the test set front panel connect one lead of an ohmmeter to J9 SCOPE, the other lead to the pins listed below. Operate the SCOPE SWITCH to the positions shown and check the circuit continuity. In each case resistance shall be less than 5 ohms.

SCOPE SWITCH position	Test signal input (connector, pin)	
1	J5	h
2	J5	w
3	J5	a

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SCOPE SWITCH position	Test signal input (connector, pin)	
4	J5	y
5	J5	q
6	J5	k
7	J5	u
8	J4	G
9	J4	E
10	J5	b
11	J5	f
12	J5	n
13	J4	y
14	J4	Z
15	J4	a
16	J4	T
17	J4	X
18	J4	Y
19	J4	n
20	J4	D
21	J4	R
*22	J4	P
23	J1	L
24	J1	P

4.7.3.2.13.2 Disconnect the ohmmeter from SCOPE jack J9 and connect it to SYNC jack J10. Measure resistance as above. In each case the resistance shall be less than 5 ohms.

SCOPE SWITCH position	Test signal input (connector, pin)	
1 thru 12	J5	y
13	J4	x
14	J4	Z

*Set XMTR CONTROL SWITCH to position 6 for this step only.

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SCOPE SWITCH position	Test signal input (connector, pin)	
15	J4	a
16	J5	y
17 thru 19	J5	h
20 thru 21	J5	y
22	OPEN	
23 and 24	J1	L

4.7.3.2.14 Pendulum signal.

4.7.3.2.14.1 Connect the equipment as shown in figure 6. Set the EXT VOLTS AC switch to position 8 and monitor the output signal for the pendulum on the ac digital voltmeter connected to the EXT VOLTS AC jack. Set the output of the 4.8 kilocycle oscillator to 25.87 to 26.13 volts rms.

4.7.3.2.14.2 With the PENDULUM control set to zero the EXT VOLTS AC output shall be less than 0.1 volt rms.

4.7.3.2.14.3 With the PENDULUM control set to CW 60 and CCW 60, the EXT VOLTS AC output signal at each setting shall be 12.0 to 12.6 volts rms.

4.7.3.2.14.4 With the PENDULUM control set to CW 10 and CCW 10, the EXT VOLTS AC output signal at each setting shall be 2.18 to 2.76 volts rms.

4.7.3.2.15 Rate gyro signal.

4.7.3.2.15.1 Set the EXT VOLTS AC switch to position 7 and monitor the output signal for the rate gyro on the ac digital voltmeter connected to the EXT VOLTS AC jack on the front panel. Determine that the output of the 4.8 kilocycle oscillator remains at 25.87 to 26.13 volts rms.

4.7.3.2.15.2 With the RATE control set to zero, the EXT VOLTS AC output shall be less than 0.1 volt rms.

4.7.3.2.15.3 With the RATE control set to CW 12 and CCW 12, the EXT VOLTS AC output indication on the ac digital voltmeter at each setting shall be 3.90 to 4.62 volts rms.

4.7.3.2.15.4 With the RATE control set to CW 30 and CCW 30, the EXT VOLTS AC output indication on the ac digital voltmeter at each setting shall be 10.3 to 11.0 volts rms.

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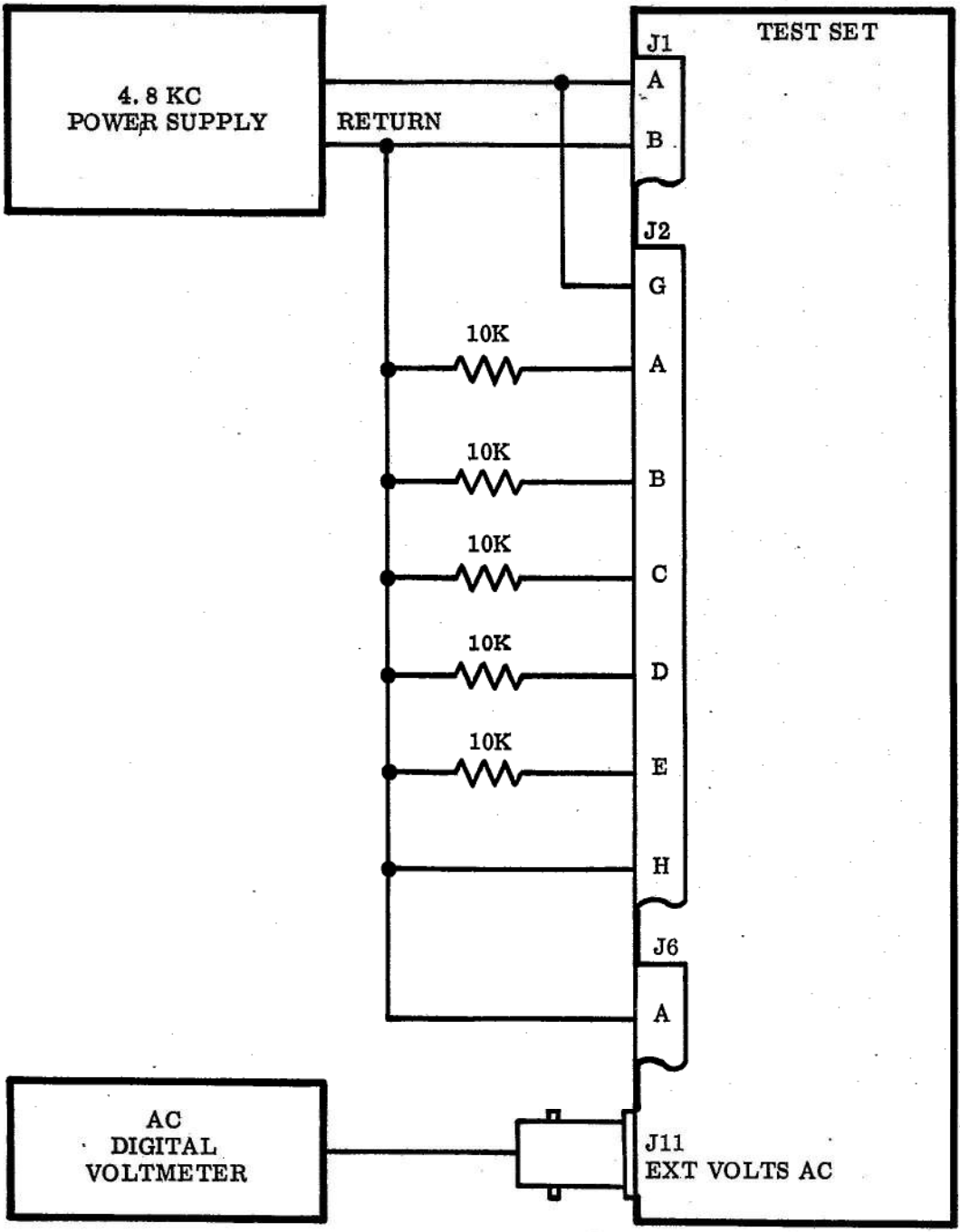


Figure 6. Rate Gyro Test

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4.7.3.2.16 Course gyro signal.

4.7.3.2.16.1 Set the EXT VOLTS AC switch to position 5 to monitor the output signal for the course gyro.

4.7.3.2.16.2 Determine that the output of the 4.8 kilocycle oscillator remains at 25.87 to 26.13 volts rms. With the COURSE GYRO control set to zero, the EXT VOLTS AC output indication shall be 12.85 to 13.15 volts rms.

4.7.3.2.16.3 With the COURSE GYRO control set to 140 degrees LEFT, the output indication shall be 22.90 to 23.50 volts rms.

4.7.3.2.16.4 With the COURSE GYRO control set to 70 degrees LEFT the output indication shall be 17.95 to 18.25 volts rms.

4.7.3.2.16.5 With the COURSE GYRO control set to 140 degrees RIGHT, the output indication shall be 2.53 to 3.13 volts rms.

4.7.3.2.16.6 With the COURSE GYRO control set to 70 degrees RIGHT, the output indication shall be 7.77 to 8.07 volts rms.

4.7.3.2.16.7 Disconnect the test equipment setup shown in figure 6.

4.7.3.2.17 Depth, level X100 signal.

4.7.3.2.17.1 Connect the equipment as shown in figure 6. Set the EXT VOLTS AC switch to position 6.

4.7.3.2.17.2 With the 4.8 kilocycle power supply level adjusted to 25.87 to 26.13 volts rms, the output of the LEVEL X100 control shall be adjustable between 0 and 22.65 to 23.57 (23.11 ±0.46) volts rms as indicated by the ac vtvm.

4.7.3.2.17.3 With the LEVEL X100 control set to 5, the ac vtvm shall indicate 11.4 to 11.7 volts rms.

4.7.3.2.17.4 Disconnect the test circuit shown in figure 6 from the test set.

4.7.3.2.18 Gyro load and phase.

4.7.3.2.18.1 Connect the test equipment as shown in figure 7. The 4.8 kc power supply output shall be 25.87 to 26.13 volts rms.

4.7.3.2.18.2 The phase meter shall indicate 18.0 to 26.0 degrees.

4.7.3.2.18.3 The ac vtvm shall indicate 275 to 305 millivolts rms.

4.7.3.2.18.4 Disconnect the test equipment.

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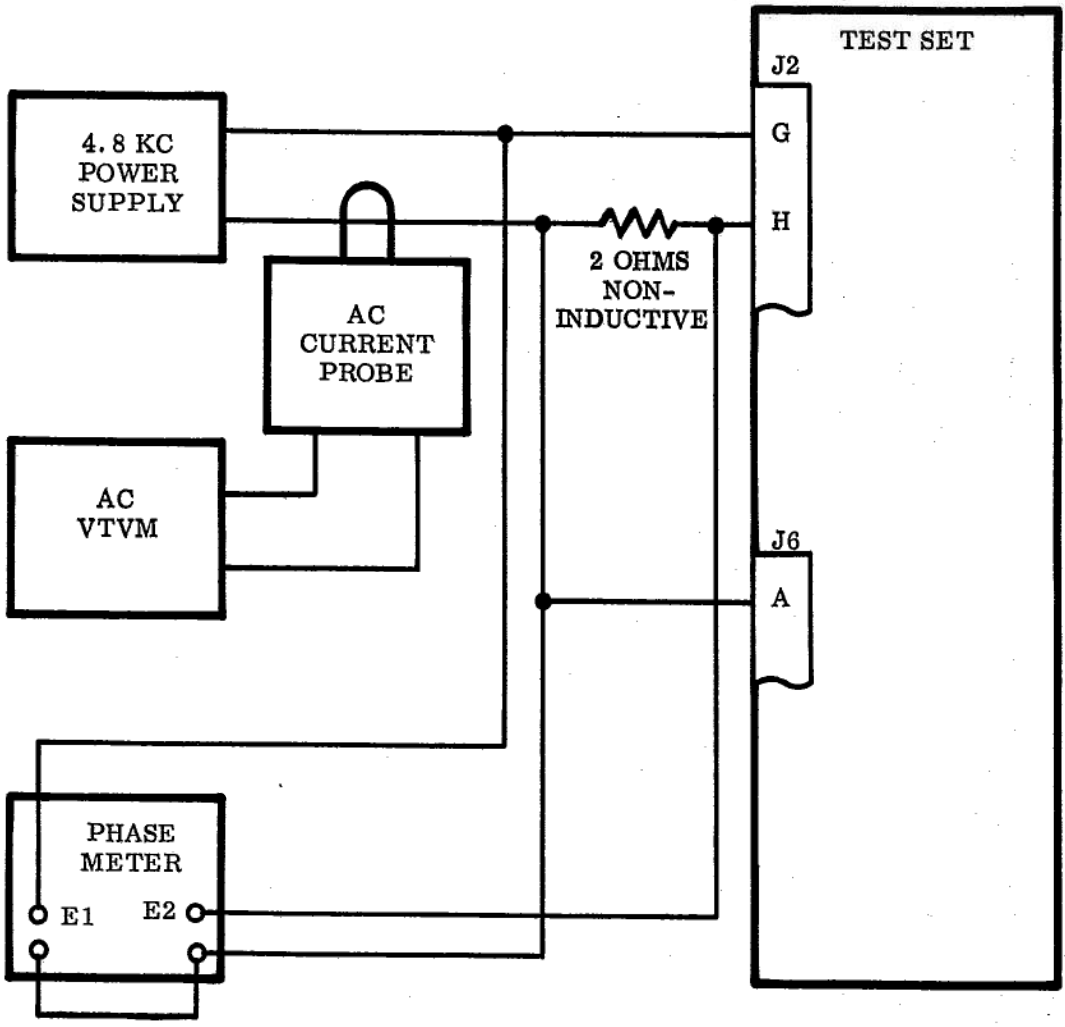


Figure 7. Course Gyroscope Load Test

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4.7.3.2.19 Memory bias.

4.7.3.2.19.1 Connect the digital dc voltmeter to J9 (SCOPE) jack. Position the TEST SET POWER switch to ON and SCOPE SWITCH to 22. The indicated voltage shall be 4.98 to 5.02 volts dc.

4.7.3.2.19.2 Connect the digital dc voltmeter to the MONITOR jacks. Position METER SWITCH to 3 and METER FUNCTION switch to 2. The indicated voltage shall be 4.98 to 5.02 volts dc.

4.7.3.2.19.3 The indication on the voltmeter on the test set front panel shall be 4.8 to 5.2 volts dc.

4.7.3.2.19.4 Connect the digital dc voltmeter to J4 pin P (-) and J6 pin A (+). Position the XMTR CONTROL SWITCH to 7. The indicated voltage shall be 4.95 to 5.05 volts dc.

4.7.3.2.19.5 Disconnect the digital dc voltmeter and position the TEST SET POWER switch to OFF.

4.7.3.2.20 Transmitter control switch.

4.7.3.2.20.1 Remove connector P3. On the connector plugs located on the test set front panel connect an ohmmeter to the pins indicated. Place the TEST SET POWER switch to ON, the TORPEDO POWER switch to ON and the FIRE CONTROL POWER switch to OFF. Operate the XMTR CONTROL SWITCH to the positions shown in table IV and check the circuit continuity. Measurements shall be as specified in table IV.

4.7.3.2.20.2 Turn the TORPEDO and TEST SET POWER switches to OFF. Replace P3. Turn the TEST SET POWER and TORPEDO POWER switches to ON. Position XMTR CONTROL SWITCH to 8. Place a jumper between J4 pin M and J6 pin Q. Connect the negative (-) lead of the dc vtvm to J6A (ground) and measure the voltage at J6 pins D, H, U, h, n, and q. In each case the indication shall be +39 to +41 (+40 ±1) volts dc.

4.7.3.2.20.3 Connect the dc vtvm positive (+) lead to J6A (ground) and measure the voltage at J6 pins S, W, Y, d, p, and v. In each case the indication shall be -39 to -41 (-40 ±1) volts dc.

4.7.3.2.20.4 Turn the TEST SET POWER and TORPEDO POWER switches to OFF. Disconnect the jumper and vtvm.

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Table IV

Transmitter Control Switch Continuity

XMTR CONTROL SWITCH Pos	Wafer A		Wafer B		Wafer C		Wafer E	
	Ohmmeter Connected between J5v and	Maximum Resistance (ohms)	Ohmmeter Connected between J4P and	Resistance (ohms)	Ohmmeter Connected between J5x and	Maximum Resistance (ohms)	Ohmmeter Connected between J5G and	Maximum Resistance (ohms)
1	J3A	5	J3A	5 max	J3A	5	J6G	5
2	J3A	5	J3A	200,000 min	J5y	5	J5a	5
3	J5w	5	J3A	200,000 min	J5V	5	J5a	5
4	J5w	5	J3A	200,000 min	J5V	5	J6G	5
5	J3A	5	J3A	5 max	J5V	5	J5a	5
6	J3A	5			J5V	5	J5a	5
7	J3A	5			J5V	5	J5a	5
8	J5w	5	J3A	5 max	J5y	5	J5a	5

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4.7.3.2.21 External volts ac switch.

4.7.3.2.21.1 On the test set front panel position the EXT VOLTS AC switch to the positions indicated and check the circuit continuity between the specified J4 pins and the EXT VOLTS AC connector. In each case the indication shall not be more than 5 ohms.

EXT VOLTS AC Switch Position	J4 Pin
3	q
4	z
9	b
10	f
11	g
12	h
13	i
14	j

4.7.3.2.21.2 On the test set front panel rotate the GYRO SELECTOR switch as indicated in table V, position the EXT VOLTS AC switch, and set the RATE control and PENDULUM controls to 0. Check circuit continuity according to table V. Resistance indications shall be as specified in Limits column.

4.7.3.2.22 400 cycle input circuit.

4.7.3.2.22.1 On the test set front panel apply a 26 vac 400 cycle signal to plug J1 pins E and D. Connect a jumper between J1 pin D and J6 pin A. Connect the ac vtvm to the FREQ COUNT jack (J8). With the COUNT SWITCH in position (400 cps), the vtvm shall indicate 3 to 8 volts rms.

4.7.3.2.22.2 Disconnect the 400 cycle supply and ac vtvm.

4.7.3.2.23 Exploder arm.

4.7.3.2.23.1 On the test set front panel place the TEST SET POWER and the TORPEDO POWER switches to ON, and the METER FUNCTION selector to position 1. The EXPLODER ARM (DS5) lamp shall be extinguished.

4.7.3.2.23.2 Connect a jumper from plug J5 pin d to plug J6 pin j. The EXPLODER ARM lamp shall be illuminated.

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Table V

External Volts ac Switch Continuity

EXT VOLTS AC Switch Position	GYRO SELECTOR Switch Position	Ohmmeter Connected Between	Limits
7	Pitch	J2D and J11*	5 ohms max
7	Pitch	J2H and J2A	5 ohms max
7	Pitch	J2H and J2B	5 ohms max
7	Pitch	J2H and J2C	0 to 1,000 ohms
7	Pitch	J2H and J2D	0 to 1,000 ohms
7	Pitch	J2H and J2E	5 ohms max
7	Roll	J2B and J11	5 ohms max
7	Roll	J2H and J2A	0 to 1,000 ohms
7	Roll	J2H and J2B	0 to 1,000 ohms
7	Roll	J2H and J2C	5 ohms max
7	Roll	J2H and J2D	5 ohms max
7	Roll	J2H and J2E	5 ohms max
7	Yaw	J2H and J11	5 ohms max
7	Yaw	J2H and J2A	0 to 1,000 ohms
7	Yaw	J2H and J2B	5 ohms max
7	Yaw	J2H and J2C	5 ohms max
7	Yaw	J2H and J2D	5 ohms max
7	Yaw	J2H and J2E	0 to 1,000 ohms
8	Pitch	J2C and J11	5 ohms max
8	Roll	J2A and J11	5 ohms max
8	Yaw	J2A and J11	5 ohms max
1	--	J60 and J11	5 ohms max
5	--	J1C and J11	5 ohms max
6	--	J2F and J11	5 ohms max
2	--	J1E and J11	5 ohms max

*J11 is the EXT VOLTS AC jack on the test set front panel. The ohmmeter plus (+) lead should be connected to the center conductor.

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CAUTION

Do not leave the METER FUNCTION switch in position 2 for more than a few seconds during the next step. After measurement, return METER FUNCTION switch to 1.

4.7.3.2.23.3 Turn the METER SWITCH to position 19. Then the METER FUNCTION switch to position 2 momentarily. The front panel voltmeter shall indicate +76 to +84 (+80 ±4) volts dc.

4.7.3.2.23.4 Return the METER FUNCTION switch to position 1. Remov the jumper and turn the TEST SET POWER switch OFF.

4.7.3.2.24 Voltmeter switch circuit.

4.7.3.2.24.1 On the test set front panel place all power controls to the OFF position and set the METER FUNCTION selector to position 1. Rotate the METER SWITCH control to the positions indicated in table VI and apply an input signal with the polarity specified to the connections shown. Use the dual dc power supply for the -15 volt power (disregard any lamp indications). The front panel meter (M1) shall indicate a reading approximately equal to the applied input voltage in each case.

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Table VI

Voltmeter Switch Circuit

METER SWITCH position	Input voltage	Input connection	Return connection
1	15	J5 V (-)	J6 G (+)
2	15	J5 Y (+)	J6 G (-)
3	5	J4 s (+)	J6 G (-)
4	30*	J5 p (+)	J6 G (-)
5	15	J5 m (-)	J6 G (+)
6	15	J5 T (-)	J6 G (+)
7	15	J5 c (-)	J6 G (+)
8	15	J5 X (-)	J6 G (+)
9	15	J5 e (-)	J6 G (+)
10	15	J5 i (-)	J6 G (+)
11	15	J5 t (-)	J6 G (+)
12	15	J5 U (-)	J6 G (+)
13	15	J5 r (-)	J6 G (+)
14	15	J5 q (-)	J6 G (+)
15	15	J5 g (-)	J6 G (+)
16	15	J5 R (-)	J6 G (+)
17	15	J6 e (-)	J6 Z (+)
18	15	J6 Z (-)	J6 e (+)
19	40*	J5 d (-)	J6 b (+)
20	40	J3 B (+)	J6 G (-)
21	40	J3 C (-)	J6 G (+)
22	30	J5 S (+)	J6 G (-)
23	30	J5 Z (-)	J6 G (+)

*Read indicated voltage on 100 volt scale.

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4.7.3.2.24.2 Place METER FUNCTION selector to position 2. Turn METER SWITCH through its 22 positions with input applied as specified in table VII. The panel meter shall indicate an output voltage approximately equal to the applied input.

Table VII

Voltmeter Switch Circuit

METER SWITCH position	Input voltage	Input connection	Return connection
1	30	J1 L (+)	J1 P (-)
4	30*	J5 L (+)	J6 G (-)
5	15	J4 c (-)	J6 G (+)
6	15	J5 W (-)	J6 G (+)
7	15	J5 P (-)	J6 G (+)
8	15	J4 C (-)	J6 G (+)
9	15	J6 J (-)	J6 G (+)
10	15	J6 E (-)	J6 G (+)
11	15	J6 M (-)	J6 G (+)
12	15	J4 r (-)	J6 G (+)
13	15	J6 T (+)	J6 G (-)
14	15	J4 d (-)	J6 G (+)
15	15**	J6 a (+)	J6 k (-)
16	15***	J6 r (+)	J6 k (-)
17	15****	J6 C (+)	J6 k (-)
18	15	J4 e (-)	J6 G (+)
20	40	J4 v (+)	J6 G (-)
21	40	J4 w (-)	J6 G (+)
22	15	J6 T (-)	J6 G (+)

*Read indicated voltage on 100 volt scale.
**TUBE-AIR/ASROC switch in TUBE position.
***CEILING MODE switch in TEST position.
****SEARCH DEPTH switch in TEST position.

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4.7.3.2.24.3 Place METER FUNCTION selector to position 3. Apply input to points specified in table VIII and turn METER SWITCH to positions listed. Indication shall be approximately 25 volts dc in each case.

Table VIII

Voltmeter Switch Circuit

METER SWITCH position	Input voltage	Input connection	Return connection
5	15	J9 (Scope jack) (-)	J6 G (+)
8	15	J5 n (-)	J6 G (+)
9	15	J5 f (-)	J6 G (+)

4.7.3.2.25 Depth function switch.

4.7.3.2.25.1 With all power switches in the OFF position, set the DEPTH FUNCTION selector to the functions indicated in table IX using J5 pin Z as a common point for the continuity check specified.

Table IX

Depth Function Switch Continuity

DEPTH FUNCTION selector position	J2 M (Wafer A)	J2 N (Wafer B)	J2 P (Wafer C)
OFF	200,000 ohms min	200,000 ohms min	200,000 ohms min
20'	5 ohms max	200,000 ohms min	200,000 ohms min
50'	5 ohms max	5 ohms max	200,000 ohms min
100'	5 ohms max	5 ohms max	5 ohms max

4.7.3.2.25.2 Measure continuity or resistance between the connector pins listed in table X with the TEST SET POWER switch in the OFF position and the METER FUNCTION switch in position 3.

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Table X

Meter Function Switch Continuity

Ref. No.	Pins		Switch Positions	Measurement
1	J4 HH	J6 V	METER SWITCH, position 16	5 ohms max
2	J4 W	J6 V	METER SWITCH, position 18	5 ohms max
3	J4 E	J6 V	METER SWITCH, position 8	5 ohms max
4	J4 E	J6 V	METER SWITCH, position 9	5 ohms max
5	J4 t	J6 V	--	approximately 8,200 ohms
6	J4 A	J6 V	--	approximately 1,000 ohms
7	J3 A	J6 k	--	approximately 4,700 ohms
8	J1 L	J1 P	--	approximately 1,000 ohms
9	J6 m	J6 a	With SQUIB FIRE pressed and TUBE-AIR/ASROC switch in the TUBE position	5 ohms max
10	J1 E	J1D	--	approximately 200 ohms
11	J2 J	J2 L	--	55 to 57 ohms

4.7.3.2.25.3 Set the TEST SET POWER switch to OFF and turn the power supply OFF. Disconnect the jumpers.

4.7.3.2.26 Alternator regulator.

4.7.3.2.26.1 Connect ohmmeter between J6 pins T and Q. Depress ALT REG load switch to IN. The ohmmeter shall indicate 19.0 to 21.0 ohms. Release ALT REG switch.

4.7.3.2.26.2 Position FUNCTION switch to 2 and METER switch to 13. Connect ohmmeter between J6 pin T and panel meter red test point. The ohmmeter shall indicate 5 ohms max.

4.7.3.2.26.3 Position METER switch to 22. Connect ohmmeter between J6 pin T and panel meter black test point. The ohmmeter shall indicate 5.0 ohms max. Disconnect ohmmeter.

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4.7.3.2.26.4 Connect a 10 ohm resistor (200 watts, ±1 percent) between J6 pins j and G. Connect a 20 ohm resistor (100 watts, ±1 percent) between J6 pins b and G. Turn on the +40 and -40 vdc power supplies. Turn the TEST SET and TORPEDO POWER switches to ON. Set the output voltage on each supply for 40 ±0.3 vdc as measured across each load resistor.

4.7.3.2.26.5 Turn the METER SWITCH to position 20, the METER FUNCTION switch to position 3. The front panel (current) meter shall indicate 1.8 to 2.2 (2.0 ±0.2) amperes.

4.7.3.2.26.6 Place the METER SWITCH to position 21. The front panel meter shall indicate 3.6 to 4.4 (4.0 ±0.4) amperes.

4.7.3.2.26.7 Turn the TEST SET and TORPEDO POWER switches to OFF, and the 40 volt power supplies to OFF. Disconnect the test circuit.

4.7.3.2.27 Yaw and pitch angle data diodes.

4.7.3.2.27.1 Connect the positive (+) terminal of the 10 volt power supply to one end of an 8,200 ohm 0.5 watt resistor. Connect the other end of the resistor to J5 pin u. Connect the negative (-) side of the 10 volt power supply to J3 pin A. Turn the TORPEDO POWER switch to OFF. Connect the dc vtvm to the SCOPE output jack on the front panel of the test set. Read positive voltage indication.

4.7.3.2.27.2 Set the SCOPE SWITCH to position 7. Set the METER SWITCH to position 1 and METER FUNCTION switch to position 3.

4.7.3.2.27.3 Turn the 10 volt power supply ON and adjust the output for 10.0 ±0.1 volts dc. The dc vtvm shall indicate 1.1 to 1.3 (1.2 ±0.1) volts dc.

4.7.3.2.27.4 Turn the METER SWITCH to position 2. The voltmeter shall indicate 9.9 to 10.1 (10.0 ±0.1) volts dc.

4.7.3.2.27.5 Disconnect the resistor from J5 pin u and reconnect it to J5 pin k. Turn the SCOPE SWITCH to position 6 and METER SWITCH to position 2. The dc vtvm shall indicate 1.1 to 1.3 (1.2 ±0.1) volts dc.

4.7.3.2.27.6 Turn the METER SWITCH to position 1. The voltmeter shall indicate 9.9 to 10.1 (10.0 ±0.1) volts dc.

4.7.3.2.27.7 Reverse the leads at the 10 volt power supply (negative to positive, positive to negative). Disconnect the resistor from J5 pin k and reconnect it to J5 pin u. Repeat steps 4.7.3.2.27.2 through 4.7.3.2.27.6 above with voltmeter set to read negative voltage.

4.7.3.2.27.8 Disconnect all power supplies and test circuits.

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4.7.3.2.28 External transmitter load.

4.7.3.2.28.1 Connect the impedance bridge between the pins on P201 of the external transmitter load as listed in the following chart. The resistance values shall be within the limits listed.

P201 Pins	Limits (Ohms)
f to h	1.15 to 1.19
j to m	1.15 to 1.19
R to P	6.44 to 6.70
T to S	6.44 to 6.70
W to Y	1.91 to 1.99
a to c	1.91 to 1.99
J to L	142.5 to 157.5
H to K	142.5 to 157.5
B to D	142.5 to 157.5
A to C	142. 5 to 157.5
n to Z	950 to 1050

4.7.3.2.28.2 Connect the ohmmeter between the following points. The resistance shall be 5 ohms maximum in each case.

Connectors and Pins

P201 Pins		Test Point
n	to	J19 (VERT BEAM CHG)
a	to	J22 (AMPL NO. 2)
f	to	J23 (AMPL NO. 4)
m	to	J24 (AMPL NO. 3)
R	to	J25 (AMPL NO. 6)
S	to	J20 (AMPL NO. 1)
Y	to	J21 (AMPL NO. 5)
c	to	P201 h
j	to	P201 h
P	to	P201 h
T	to	P201 h
W	to	P201 h
d	to	P201 U
i	to	J26 P

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4.7.4 Examination of product. The test set shall be examined for conformance to the requirements specified herein and on the drawings listed in BuWeps DL 2540800. The presence of any one of the following defects shall constitute a failure:

4.7.4.1 Test set not complete; components or parts missing.

4.7.4.2 Component parts not new, evidence of reclaimed, used, or salvaged components.

4.7.4.3 Evidence of unauthorized materials or repairs.

4.7.4.4 Receptacles do not accept mating plugs.

4.7.4.5 Panel inscriptions incomplete, incorrect, or illegible.

4.7.4.6 Case not watertight.

4.7.4.7 Continuity not in accordance with schematic.

4.7.4.8 Test set not correctly identified.

4.7.4.9 Test set not preserved and packaged in accordance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Preservation, and packaging of the test set for level A shipment shall be in accordance with Specification MIL-P-116. Cleaning shall be accomplished in accordance with Process C-1 and drying, if required shall be accomplished in accordance with Procedure D-1. Equipment case shall serve as rigid metal container.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation and packaging of the test set shall be in accordance with the supplier's own commercial practice, when such practices have been proven by testing to conform with all applicable requirements.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. The test set, preserved and packaged in accordance with 5.1, shall be overpacked in containers in compliance with Specification PPP-B-636 Type CF, Class 2, weather resistant.

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5.2.2 Level B. Not applicable.

5.2.3 Level C. The test set, when preserved and packaged as specified in 5.1, shall be overpacked to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with I.C.C. Regulation, and with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2.4 Cushioning or dunnage. Supplier shall provide sufficient cushioning or dunnage material to protect equipment adequately from damage due to shock or vibration loads encountered during handling and transportation, and to assure safe delivery to the receiving activity.

5.3 Marking.

5.3.1 Normal marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Standard MIL-STD-129.

6. NOTES

6.1 Description and intended use.

6.1.1 Description of test set. The test set is a portable unit of electronic test equipment contained in a watertight, shock protected combination case with a removable cover and facilities for storage of special tools and cabling. The case also provides protection from environmental deterioration while in storage and damage from handling during shipment. The test set front panel provides controls and connectors accessible for operation and interconnection of the test set when the cover is removed. The test set consists of electronic circuitry which provides proper simulated signals and loads to the Guidance Control group being tested.

6.1.2 Intended use. The test set is intended for use in the fleet or at a manufacturing facility for verifying the proper performance of the guidance and control systems for Torpedoes Mk 46 Mod 0 and Mk 46 Mod 1 in accordance with the following documents:

Mk 46 Mod 0	OP 2656
Mk 46 Mod 1	OP 3389 MIL-T-82426

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Required level of preservation and packaging.

(c) Required level of packing.

(d) Assigned activity for preproduction and periodic tests.

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(e) Required data.

(f) Any exceptions to this specification.

(g) All additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.3 Definitions.

6.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement is one half the peak-to-peak excursion in inches or the corresponding maximum acceleration in gravity units (g). The frequency is given in cycles per second (cps).

6.3.2 Shock. Shock is an abrupt change in the position of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse force in units of gravitational acceleration (g's), of duration in units of time, milliseconds (ms), and a shock wave approximating a one-half sine wave. The time duration shall be measured between two points where two lines intersect the zero "g" line. These lines are tangent to the shock wave slopes.

6.3.3 Temperature. The temperature of the environment to which the equipment may be exposed, expressed in degrees Fahrenheit (° F).

6.3.4 Humidity. The relative humidity of the environment to which the test set is exposed, expressed as a percent.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level B Not applicable.

Level C This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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REPORT FORM

QUALITY ASSURANCE PROVISION

ACCEPTANCE TESTS AND EXAMINATIONS

FOR

GUIDANCE CONTROL TEST SET

DL 2540800

(For Torpedo Mk 46 Mod 0 and Mod 1)

EQUIPMENT STATUS

CONTRACTOR CERTIFICATION		GOVERNMENT ACCEPTANCE	
Contractor		Date of Acceptance	
Contract		Accepted by	
Equipment Ser. No.		Title	
Insp. Req. Satisfied		Agency	

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REPORT FORM

QUALITY ASSURANCE PROVISION

ACCEPTANCE TESTS AND EXAMINATIONS

GUIDANCE CONTROL TEST SET DL 2540800

10. SCOPE

10.1 This Report Form contains data sheets to record the results of the acceptance tests and examinations of 4.7 of this specification. When completed, with authorized certification and acceptance signatures affixed, this Report Form, or copies thereof, shall accompany the test set bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test Sequence. The acceptance tests and examinations of 4.7 shall be completed and the results recorded or verified in this Report Form prior to acceptance of the test set. Unless otherwise noted the sequence of testing, recording, and verification, as performed in 4.7, is mandatory and must be performed in the order shown.

20.2 Recording of Results. Where space is provided to indicate the value of specific reading obtained, the specific reading shall be recorded if a limit or limits is given in the Limits column for the test. The initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated in the Limits column, the individual performing the test or inspection shall initial in the space provided in the Reading column indicating that the requirement has been met satisfactorily.

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30. ACCEPTANCE TEST AND EXAMINATION

	Reference	Limits	Reading
30.1 Test Equipment	4.7.2	In calibration	_____
30.2 Test Procedure	4.7.3		
30.2.1 Test Setup	4.7.3.1		
30.2.1.1 Guide Pin Orientation	4.7.3.1.1		
J17 (F) next to pin A		Correctly oriented	_____
P17 (M) next to pin A		Correctly oriented	_____
J2350 (M) next to pin A		Correctly oriented	_____
J14 (M) next to pin A		Correctly oriented	_____
P14 (F) next to pin A		Correctly oriented	_____
J2351 (M) next to pin A		Correctly oriented	_____
J18 (M) next to pin A		Correctly oriented	_____
P18 (F) next to pin A		Correctly oriented	_____
P2302-A (F) next to pin A		Correctly oriented	_____
P2302-B (F) next to pin A		Correctly oriented	_____
J28 (F) next to pin 1		Correctly oriented	_____
P28 (M) next to pin 1		Correctly oriented	_____
P5511 (M) next to pin 1		Correctly oriented	_____
J15 (F) next to pin 1		Correctly oriented	_____
P15 (M) next to pin 1		Correctly oriented	_____
P2323 (F) next to pin A		Correctly oriented	_____
P2324 (F) next to pin A		Correctly oriented	_____
P3602 (F) next to pin A		Correctly oriented	_____
J4202 (F) next to pin A		Correctly oriented	_____
30.2.1.2 Continuity Check	4.7.3.1.2		
P2323 - A to P4 - W		5 ohms max	_____
B X		5 ohms max	_____
C Y		5 ohms max	_____
D Z		5 ohms max	_____
P2323 - E to P4 a		5 ohms max	_____

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30.2.1.2 (Cont)

		Reference	Limits	Reading
P2323 - F	to P4 - b		5 ohms max	
H	c		5 ohms max	
J	d		5 ohms max	
K	e		5 ohms max	
L	f		5 ohms max	
M	g		5 ohms max	
N	h		5 ohms max	
P	i		5 ohms max	
R	j		5 ohms max	
T	m		5 ohms max	
U	n		5 ohms max	
P2323 - V	to P4 - p		5 ohms max	
P2324 - C	to P4 - HH		5 ohms max	
E	q		5 ohms max	
F	r		5 ohms max	
H	GG		5 ohms max	
J	s		5 ohms max	
N	t		5 ohms max	
P	u		5 ohms max	
R	v		5 ohms max	
S	w		5 ohms max	
T	x		5 ohms max	
U	y		5 ohms max	
V	EE		5 ohms max	
P2324 - X	to P4 - z		5 ohms max	
P2302A -A	to P4 - A		5 ohms max	
C	C		5 ohms max	
D	D		5 ohms max	
E	E		5 ohms max	
H	G		5 ohms max	
J	H		5 ohms max	
P2302A -K	to P4 - J		5 ohms max	

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30.2.1.2 (Cont)				Reference	Limits	Reading
P2302A	-	P	to	M	5 ohms max	
		R		N	5 ohms max	
		S		P	5 ohms max	
		T		R	5 ohms max	
P2302A	-	V	to	T	5 ohms max	
J2351	-	A	to	P2350 - A	5 ohms max	
		B		P2350 - B	5 ohms max	
		B		P5 - u	5 ohms max	
		C		P5 - t	5 ohms max	
		E		P5 - s	5 ohms max	
		F		P2350 - F	5 ohms max	
		F		P5 - r	5 ohms max	
		J		P2350 - J	5 ohms max	
		J		P5 - q	5 ohms max	
		K		P5 - y	5 ohms max	
		L		P2350 - L	5 ohms max	
		L		P5 - n	5 ohms max	
		M		P2350 - M	5 ohms max	
		N		P2350 - N	5 ohms max	
		P		P2350 - P	5 ohms max	
		P		P5 - m	5 ohms max	
		R		P2350 - R	5 ohms max	
		S		P2350 - S	5 ohms max	
		S		P5 - k	5 ohms max	
		T		P2350 - T	5 ohms max	
		T		P5 - i	5 ohms max	
		U		P2350 - U	5 ohms max	
		U		P5 - h	5 ohms max	
		V		P2350 - V	5 ohms max	
		W		P2350 - W	5 ohms max	
		W		P5 - g	5 ohms max	
		X		P2350 - X	5 ohms max	
J2351	-	X	to	P5 - f	5 ohms max	

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30.2.1.2 (Cont)			Reference	Limits	Reading
J2351	- Z	to	P2350 - Z	5 ohms max	_____
	Z		P5 - e	5 ohms max	_____
	b		P5 - j	5 ohms max	_____
	b		P5 - d	5 ohms max	_____
	d		P5 - p	5 ohms max	_____
	f		P2350 - f	5 ohms max	_____
	g		P5 - L	5 ohms max	_____
	h		P2350 - h	5 ohms max	_____
	h		P5 - c	5 ohms max	_____
	i		P2350 - i	5 ohms max	_____
	j		P2350 - j	5 ohms max	_____
	k		P2350 - k	5 ohms max	_____
	m		P2350 - m	5 ohms max	_____
	n		P2350 - n	5 ohms max	_____
	n		P5 - b	5 ohms max	_____
	p		P5 - a	5 ohms max	_____
	q		P2350 - q	5 ohms max	_____
	q		P5 - Z	5 ohms max	_____
	s		P2350 - s	5 ohms max	_____
	s		P5 - Y	5 ohms max	_____
	t		P2350 - t	5 ohms max	_____
	t		P5 - X	5 ohms max	_____
	u		P2350 - u	5 ohms max	_____
	v		P2350 - v	5 ohms max	_____
	x		P5 - w	5 ohms max	_____
	y		P2350 - y	5 ohms max	_____
	z		P2350 - z	5 ohms max	_____
	z		P2302B - S	5 ohms max	_____
	AA		P2350 - AA	5 ohms max	_____
	AA		P5 - W	5 ohms max	_____
	AB		P2350 - AB	5 ohms max	_____
J2351	- AC	to	P2350 - AC	5 ohms max	_____

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30.2.1.2 (Cont)	Reference	Limits	Reading
J2351 - AE to P2350 - AE		5 ohms max	_____
AE P5 - V		5 ohms max	_____
AF P2350 - AF		5 ohms max	_____
AF P5 - U		5 ohms max	_____
AH P2350 - AH		5 ohms max	_____
AH P5 - T		5 ohms max	_____
AJ P2350 - AJ		5 ohms max	_____
AN P2350 - AN		5 ohms max	_____
AN P5 - S		5 ohms max	_____
AP P2350 - AP		5 ohms max	_____
J2351 - AP to P5 - P		5 ohms max	_____
J2350 - K to P5 - x		5 ohms max	_____
x v		5 ohms max	_____
d H		5 ohms max	_____
J2350 - p to P5 - G		5 ohms max	_____
P5511 - 1 to P6 - A		5 ohms max	_____
3 C		5 ohms max	_____
5 D		5 ohms max	_____
9 F		5 ohms max	_____
10 G		5 ohms max	_____
11 H		5 ohms max	_____
15 K		5 ohms max	_____
29 S		5 ohms max	_____
30 L		5 ohms max	_____
34 T		5 ohms max	_____
35 U		5 ohms max	_____
36 W		5 ohms max	_____
40 X		5 ohms max	_____
41 Y		5 ohms max	_____
45 a		5 ohms max	_____
47 d		5 ohms max	_____
P5511 - 50 to P6 - f		5 ohms max	_____

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30.2.1.2 (Cont)	Reference	Limits	Reading
P5511 - 51 to P6 - g		5 ohms max	
52 h		5 ohms max	
56 k		5 ohms max	
57 m		5 ohms max	
58 n		5 ohms max	
59 p		5 ohms max	
60 q		5 ohms max	
62 r		5 ohms max	
63 s		5 ohms max	
P5511 - 65 v		5 ohms max	
Terminal Lug V		5 ohms max	
P5511 - 28 P		5 ohms max	
22 O		5 ohms max	
38 N		5 ohms max	
P5511 - 23 to P6 - Q		5 ohms max	
J4202 - L to P2 - A		5 ohms max	
U B		5 ohms max	
T C		5 ohms max	
S D		5 ohms max	
K E		5 ohms max	
P F		5 ohms max	
D G		5 ohms max	
J H		5 ohms max	
A J		5 ohms max	
C K		5 ohms max	
B L		5 ohms max	
R M		5 ohms max	
M N		5 ohms max	
F P		5 ohms max	
E R		5 ohms max	
J4202 - W to P2 - S		5 ohms max	

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		Reference	Limits	Reading
30.2.1.2 (Cont)				
	P3602 - A to P1 - A		5 ohms max	_____
	B B		5 ohms max	_____
	C C		5 ohms max	_____
	D D		5 ohms max	_____
	E E		5 ohms max	_____
	J J		5 ohms max	_____
	H H		5 ohms max	_____
	P P		5 ohms max	_____
	P3602 - L to P1 - L		5 ohms max	_____
30.2.1.3	Power Supply Outputs	4.7.3.1.3		
	+40 volt supply		+39 to +41 volts dc	_____
	-40 volt supply		-39 to -41 volts dc	_____
30.2.2	Performance Tests (Simulated)	4.7.3.2		
30.2.2.1	Over Voltage Protection	4.7.3.2.1		
	TEST SET POWER lamp:			
	At turn-on	4.7.3.2.1.2	Illuminates	_____
	+ voltage increased	4.7.3.2.1.3	Extinguishes	_____
	Extinguishing voltage		+42.0 to +44.0 vdc	_____
	ALT REG LOAD switch IN	4.7.3.2.1.4	Illuminates	_____
	- voltage increased	4.7.3.2.1.5	Extinguishes	_____
	Extinguishing voltage		-42.0 to -44.0 vdc	_____
	ALT REG LOAD switch IN	4.7.3.2.1.6	Illuminates	_____

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		Reference	Limits	Reading
30.2.2.2	Bistable Multivibrator Circuit	4.7.3.2.2		
	Voltage levels measured at J7 acc. Table III	4.7.3.2.2.2		

Step No.	Test Ckt Switch Pos.	Limits	Reading
1	S3, P & R	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____
2	S3, P & R	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____
3	S3, P	20 to 25 volts	_____
	S3, R	0 to 2 volts	_____
4	S3, P & R	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____
5	TORP PWR OFF, then ON	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____
6	TORP PWR OFF, then ON	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____
7	S3, P & R	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____
8	S3, P & R	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____
9	S3, P	20 to 25 volts	_____
	S3, R	0 to 2 volts	_____
10	S3, P & R	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____
11	S3, P & R	20 to 25 volts	_____
	S4, P & R	0 to 2 volts	_____

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30.2.2.2 (Cont)	Step No.	Test Ckt Switch Pos.	Reference	Limits	Reading
	12	S3, P & R		20 to 25 volts	
		S4, P & R		0 to 2 volts	
	13	S3, P & R		20 to 25 volts	
		S4, P & R		0 to 2 volts	
	14	S3, P		20 to 25 volts	
		S3, R		0 to 2 volts	
	15	S3, P		20 to 25 volts	
		S3, R		0 to 2 volts	
	16	S3, P		20 to 25 volts	
		S3, R		0 to 2 volts	
	17	S3, P & R		20 to 25 volts	
		S4, P & R		0 to 2 volts	
30.2.2.3	Bistable Reset		4.7.3.2.3		
	Voltage indication		4.7.3.2.3.2	20 to 25 vdc	
	Voltage indication		4.7.3.2.3.3	-0.5 to +0.5 vdc	
30.2.2.4	Diode Steering		4.7.3.2.4		
	Voltage change, J5-q to J6-A (GRD)		4.7.3.2.4.3	From approx. 1 to 25 volts p-p at 0.5 cps	
	Voltage change, J5-w to J6-A (GRD)		4.7.3.2.4.4	From approx. 1 to 25 volts p-p at 0.5 cps	

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		Reference	Limits	Reading
30.2.2.5	Oscillator Amplifier Speed Control Circuit	4.7.3.2.5		
	ALT FREQ at maximum increase	4.7.3.2.5.2	Greater than 2240 cps	_____
	ALT FREQ at maximum decrease	4.7.3.2.5.3	1200 to 1700 cps	_____
	Oscillator Output	4.7.3.2.5.4	28 to 36 volts rms	_____
30.2.2.6	Fire Control Regulator Circuit	4.7.3.2.6		
	Maximum fire control voltage (panel meter)	4.7.3.2.6.4	30 to 33 volts dc	_____
	Minimum fire control voltage indication	4.7.3.2.6.5	17 to 20 volts dc	_____
	Current limit	4.7.3.2.6.7	6 to 8 amperes	_____
30.2.2.7	Fire Control (Search Depth)	4.7.3.2.7		
	SEARCH DEPTH lamp, fire control power applied	4.7.3.2.7.3	SD lamp extinguished	_____
	Search depth voltage at J6 pins k and F	4.7.3.2.7.4	30 to 33 volts dc	_____
	SEARCH DEPTH switch			
	Position 2	4.7.3.2.7.5	SD lamp extinguished	_____
	Position 3	4.7.3.2.7.6	SD lamp extinguished	_____
	Position 4	4.7.3.2.7.7	SD lamp extinguished	_____

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		Reference	Limits	Reading
30.2.2.7	(Cont)			
	Position 5	4.7.3.2.7.8	SD lamp extinguished	_____
	Position 6	4.7.3.2.7.9	SD lamp extinguished	_____
	Position 7	4.7.3.2.7.10	SD lamp illuminates	_____
	TEST position	4.7.3.2.7.11	SD lamp illuminated	_____
	Fire control voltage	4.7.3.2.7.12	20 to 30 volts	_____
30.2.2.8	Fire Control (Ceiling Mode	4.7.3.2.8		
	CEILING MODE lamp	4.7.3.2.8.2	Extinguished	_____
	Ceiling mode voltage at J6 pins k and K	4.7.3.2.8.3	30 to 33 volts	_____
	CEILING MODE switch			
	Position 2	4.7.3.2.8.4	CM lamp extinguished	_____
	Position 3	4.7.3.2.8.5	CM lamp extinguished	_____
	Position 4	4.7.3.2.8.6	CM lamp extinguished	_____
	Position 5	4.7.3.2.8.7	CM lamp illuminated	_____
	TEST position	4.7.3.2.8.8	CM lamp illuminated	_____
	Fire Control Voltage		25 to 35 volts	_____
30.2.2.9	Light Circuits	4.7.3.2.9		
	RIGHT, CW, STEERING COMMAND lamps	4.7.3.2.9.2	Illuminated	_____
	Panel meter indication	4.7.3.2.9.3	37 to 43 volts dc	_____
	DOWN lamp	4.7.3.2.9.4	Extinguished	_____
	UP lamp		illuminated	_____

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		Reference	Limits	Reading
30.2.2.9	(Cont)			
	Panel meter		-0.5 to +0.5 volt dc	_____
	Panel meter	4.7.3.2.9.5	37 to 43 volts dc	_____
	STEERING COMMAND lamps	4.7.3.2.9.6		
	CW ROLL		Extinguished	_____
	CCW ROLL		Illuminated	_____
	Panel meter		-0.5 to +0.5 volt dc	_____
	Panel meter	4.7.3.2.9.7	37 to 43 volts dc	_____
	STEERING COMMAND lamps	4.7.3.2.9.8		
	RIGHT		Extinguished	_____
	LEFT		Illuminated	_____
	Panel meter		-0.5 to +0.5 volt dc	_____
	SPEED CONTROL INCREASE lamp	4.7.3.2.9.11	Illuminated	_____
		4.7.3.2.9.12	Remains illuminated	_____
	SPEED CONTROL DECREASE lamp	4.7.3.2.9.13	Illuminated	_____
		4.7.3.2.9.14	Remains Illuminated	_____
	Distribution box MOD 0 lamp	4.7.3.2.9.15	Illuminated	_____
	MOD 0 push-to-test lamp depressed		Illuminated	_____
	MOD 1 push-to-test lamp depressed		Illuminated	_____

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		Reference	Limits	Reading
30.2.2.9	(Cont)			
	J28 voltages	4.7.3.2.9.16		
	Pin 53		-38.5 to -41.5 volts dc	_____
	Pin 24		-38.5 to -41.5 volts dc	_____
	Pin 46		+38.5 to +41.5 volts dc	_____
	Steering lamps	4.7.3.2.9.17		
	RIGHT lamp		Illuminated	_____
	CW lamp		Illuminated	_____
	DOWN lamp		Illuminated	_____
	J28 pins 10 and 12 jumpered	4.7.3.2.9.18		
	DOWN lamp		Extinguished	_____
	UP lamp		Illuminated	_____
	J28 pins 10 and 6 jumpered	4.7.3.2.9.19		
	CW lamp		Extinguished	_____
	CCW lamp		Illuminated	_____
	J28 pins 10 and 18 jumpered	4.7.3.2.9.20		
	RIGHT lamp		Extinguished	_____
	LEFT lamp		Illuminated	_____
	J28 pins 23 and 39 jumpered	4.7.3.2.9.21		
	MOD 0 lamp		Extinguished	_____
	MOD 1 lamp		Illuminated	_____
	Voltage at J28, pins:	4.7.3.2.9.22		
	54		+38.5 to +41.5 volts dc	_____
	48		+38.5 to +41.5 volts dc	_____

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		Reference	Limits	Reading
30.2.2.9	(Cont)			
	46		+38.5 to +41.5 volts dc	_____
	24	4.7.3.2.9.23	-38.5 to -41.5 volts dc	_____
	49		-38.5 to -41.5 volts dc	_____
	53		-38.5 to -41.5 volts dc	_____
	55		-38.5 to -41.5 volts dc	_____
	Continuity, J14, pin D to J12, pin n	4.7.3.2.9.24	5 ohms max	_____
	J28, pins 10 and 18, 53 and 16 jumpered	4.7.3.2.9.25		
	LEFT lamp		Extinguished	_____
	RIGHT lamp		Illuminated	_____
	J28, pins 16 and 46 jumpered	4.7.3.2.9.26		
	RIGHT lamp		Extinguished	_____
	LEFT lamp		Illuminated	_____
	J28, pins 10 and 42, 46 and 43 jumpered	4.7.3.2.9.27		
	UP lamp		Extinguished	_____
	DOWN lamp		Illuminated	_____
	J28, pins 43 and 53 jumpered	4.7.3.2.9.28		
	DOWN lamp		Extinguished	_____
	UP lamp		Illuminated	_____
	S1 position:	4.7.3.2.9.29		
	1			
	CCW lamp		Extinguished	_____
	CW lamp		Illuminated	_____

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		Reference	Limits	Reading
30.2.2.9	(Cont)			
	S1 position: (Cont)			
	2			
	CW lamp		Extinguished	_____
	CCW lamp		Illuminated	_____
	3			
	CW lamp		Extinguished	_____
	CCW lamp		Illuminated	_____
	4			
	CCW lamp		Extinguished	_____
	CW lamp		Illuminated	_____
30.2.2.10	Gyro Relay Circuit	4.7.3.2.10		
	Resistance at J4, pin m and J6, pin Q	4.7.3.2.10.2	Greater than 200,000 ohms	_____
	Resistance with METER switch at 1	4.7.3.2.10.3	Less than 5 ohms	_____
30.2.2.11	Circuit Continuity	4.7.3.2.11		
	SYSTEM CONTINUITY lamp	4.7.3.2.11.1	Extinguished	_____
	SYSTEM CONTINUITY lamp with J6, pins s and g jumpered	4.7.3.2.11.2	Illuminated	_____
	Resistance, J2, pins R and S	4.7.3.2.11.3	Less than 5 ohms	_____
	Resistance, J1, pins J and H	4.7.3.2.11.4	Less than 5 ohms	_____
30.2.2.12	Inverter Circuit	4.7.3.2.12		
	Oscilloscope traces	4.7.3.2.12.3	In phase	_____
		4.7.3.2.12.4	Out of phase	_____

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		Reference	Limits	Reading
30.2.2.13	Scope Switch Continuity	4.7.3.2.13		
	Scope J9	4.7.3.2.13.1		

Scope Switch Position		
1	5 ohms max	_____
2	5 ohms max	_____
3	5 ohms max	_____
4	5 ohms max	_____
5	5 ohms max	_____
6	5 ohms max	_____
7	5 ohms max	_____
8	5 ohms max	_____
9	5 ohms max	_____
10	5 ohms max	_____
11	5 ohms max	_____
12	5 ohms max	_____
13	5 ohms max	_____
14	5 ohms max	_____
15	5 ohms max	_____
16	5 ohms max	_____
17	5 ohms max	_____
18	5 ohms max	_____
19	5 ohms max	_____
20	5 ohms max	_____
21	5 ohms max	_____
22	5 ohms max	_____
23	5 ohms max	_____
24	5 ohms max	_____

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	Reference	Limits	Reading
30.2.2.13 (Cont)			
SYNC J10	4.7.3.2.13.2		
Scope Switch Position			
1		5 ohms max	
2		5 ohms max	
3		5 ohms max	
4		5 ohms max	
5		5 ohms max	
6		5 ohms max	
7		5 ohms max	
8		5 ohms max	
9		5 ohms max	
10		5 ohms max	
11		5 ohms max	
12		5 ohms max	
13		5 ohms max	
14		5 ohms max	
15		5 ohms max	
16		5 ohms max	
17		5 ohms max	
18		5 ohms max	
19		5 ohms max	
20		5 ohms max	
21		5 ohms max	
22		Greater than 200,000 ohms	
23		5 ohms max	
24		5 ohms max	

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		Reference	Limits	Reading
30.2.2.14	Pendulum Gyro Signal	4.7.3.2.14		
	EXT VOLTS AC output:			
	with PENDULUM control at 0	4.7.3.2.14.2	Less than 0.1 volt rms	_____
	with PENDULUM control at CW 60 and CCW 60	4.7.3.2.14.3	12.0 to 12.6 volts rms	___CW ___CCW
	with PENDULUM control at CW 10 and CCW 10	4.7.3.2.14.4	2.18 to 2.76 volts rms	___CW ___CCW
30.2.2.15	Rate Gyro Signal	4.7.3.2.15		
	EXT VOLTS AC output			
	RATE control at zero	4.7.3.2.15.2	Less than 0.1 volt rms	_____
	RATE control at CW12 and CCW12	4.7.3.2.15.3	3.90 to 4.62 volts rms	___CW ___CCW
	RATE control at CW30 and CCW30	4.7.3.2.15.4	10.3 to 11.0 volts rms	___CW ___CCW
30.2.2.16	Course Gyro Signal	4.7.3.2.16		
	Output (EXT VOLTS AC) with COURSE GYRO control set to:			
	Zero	4.7.3.2.16.2	12.85 to 13.15 volts rms	_____

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		Reference	Limits	Reading
30.2.2.16	(Cont)			
	140 degrees left	4.7.3.2.16.3	22.90 to 23.50 volts rms	_____
	70 degrees left	4.7.3.2.16.4	17.95 to 18.25 volts rms	_____
	140 degrees right	4.7.3.2.16.5	2.53 to 3.13 volts rms	_____
	70 degrees right	4.7.3.2.16.6	7.77 to 8.07 volts rms	_____
30.2.2.17	Depth Level x 100 Signal	4.7.3.2.17		
	LEVEL X100 control range	4.7.3.2.17.2	Adjustable from 0 to 22.65 to 23.57 volts rms	_____
	Output with LEVEL X100 set at 5	4.7.3.2.17.3	11.4 to 11.7 volts rms	_____
30.2.2.18	Gyro Load and Phase	4.7.3.2.18		
	Phase meter indication	4.7.3.2.18.2	18.0 to 26.0 degrees	_____
	Ac vtvm	4.7.3.2.18.3	275 to 305 millivolts rms	_____
30.2.2.19	Memory Bias	4.7.3.2.19		
	SCOPE SWITCH on 22	4.7.3.2.19.1	Vtvm indicates 4.98 to 5.02 volts dc	_____
	METER SWITCH on 3	4.7.3.2.19.2	Vtvm indicates 4.98 to 5.02 volts dc	_____

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		Reference	Limits	Reading
30.2.2.19	(Cont)			
	Panel meter indication	4.7.3.2.19.3	4.8 to 5.2 volts dc	_____
	Dig. dc voltmeter ind.	4.7.3.2.19.4	4.95 to 5.05 volts dc	_____
30.2.2.20	Transmitter Control Switch	4.7.3.2.20		
	S14 positions	4.7.3.2.20.1		
	Wafer A			
	1		5 ohms max	_____
	2		5 ohms max	_____
	3		5 ohms max	_____
	4		5 ohms max	_____
	5		5 ohms max	_____
	6		5 ohms max	_____
	7		5 ohms max	_____
	8		5 ohms max	_____
	Wafer B			
	1		5 ohms max	_____
	2		200,000 ohms min	_____
	3		200,000 ohms min	_____
	4		200,000 ohms min	_____
	5		5 ohms max	_____
	8		5 ohms max	_____

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	Reference	Limits	Reading
30.2.2.20 (Cont)			
Wafer C			
1		5 ohms max	_____
2		5 ohms max	_____
3		5 ohms max	_____
4		5 ohms max	_____
5		5 ohms max	_____
6		5 ohms max	_____
7		5 ohms max	_____
8		5 ohms max	_____
Wafer E			
1		5 ohms max	_____
2		5 ohms max	_____
3		5 ohms max	_____
4		5 ohms max	_____
5		5 ohms max	_____
6		5 ohms max	_____
7		5 ohms max	_____
8		5 ohms max	_____
J6 pin voltages, positive	4.7.3.2.20.2		
Pins			
D		+39 to +41 volts dc	_____
H		+39 to +41 volts dc	_____
U		+39 to +41 volts dc	_____
h		+39 to +41 volts dc	_____
n		+39 to +41 volts dc	_____
q		+39 to +41 volts dc	_____

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		Reference	Limits	Reading
30.2.2.20	(Cont)			
	J6 pin voltages, negative	4.7.3.2.20.3		
	Pins			
	S		-39 to -41 volts dc	_____
	W		-39 to -41 volts dc	_____
	Y		-39 to -41 volts dc	_____
	d		-39 to -41 volts dc	_____
	p		-39 to -41 volts dc	_____
	v		-39 to -41 volts dc	_____
30.2.2.21	External Volts AC Switch	4.7.3.2.21		
	Switch Position	J4 Pin		
	3	q	5 ohms max	_____
	4	z	5 ohms max	_____
	9	b	5 ohms max	_____
	10	f	5 ohms max	_____
	11	g	5 ohms max	_____
	12	h	5 ohms max	_____
	13	i	5 ohms max	_____
	14	j	5 ohms max	_____

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30.2.2.21 (Cont)

EXT VOLTS AC Switch Continuity — Reference: 4.7.3.2.21.2

EXT VOLTS AC sw. Position	GYRO SEL. sw. Position	Ohmmeter Connected to	Limits	Reading
7	Pitch	J2D and J11*	5 ohms max	_____
7	Pitch	J2H and J2A	5 ohms max	_____
7	Pitch	J2H and J2B	5 ohms max	_____
7	Pitch	J2H and J2C	0 to 1,000 ohms	_____
7	Pitch	J2H and J2D	0 to 1,000 ohms	_____
7	Pitch	J2H and J2E	5 ohms max	_____
7	Roll	J2B and J11	5 ohms max	_____
7	Roll	J2H and J2A	0 to 1,000 ohms	_____
7	Roll	J2H and J2B	0 to 1,000 ohms	_____
7	Roll	J2H and J2C	5 ohms max	_____
7	Roll	J2H and J2D	5 ohms max	_____
7	Roll	J2H and J2E	5 ohms max	_____
7	Yaw	J2E and J11	5 ohms max	_____
7	Yaw	J2H and J2A	0 to 1,000 ohms	_____
7	Yaw	J2H and J2B	5 ohms max	_____
7	Yaw	J2H and J2C	5 ohms max	_____
7	Yaw	J2H and J2D	5 ohms max	_____
7	Yaw	J2H and J2E	0 to 1,000 ohms	_____
8	Pitch	J2C and J11	5 ohms max	_____
8	Roll	J2A and J11	5 ohms max	_____
8	Yaw	J2A and J11	5 ohms max	_____

*Ref. footnote, test, table V, p. 46

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			Reference	Limits	Reading
30.2.2.21	(Cont)				
EXT VOLTS AC sw. Position	GYRO SEL. sw. Position	Ohmmeter Connected to			
1	---	J60 and J11		5 ohms max	_____
5	---	J1C and J11		5 ohms max	_____
6	---	J2F and J11		5 ohms max	_____
2	---	J1E and J11		5 ohms max	_____
30.2.2.22	400 Cycle Input Circuit		4.7.3.2.22		
	Ac vtvm indication		4.7.3.2.22.1	3 to 8 volts rms	_____
30.2.2.23	Exploder Arm		4.7.3.2.23		
	EXPLODER ARM lamp		4.7.3.2.23.1	Extinguished	_____
	EXPLODER ARM lamp		4.7.3.2.23.2	Illuminated	_____
	Panel meter M1 indication		4.7.3.2.23.3	+76 to +84 volts dc	_____
30.2.2.24	Voltmeter Switch Circuit		4.7.3.2.24		
	Approximate panel meter M1 indications with METER FUNCTION switch in position 1, METER SWITCH in following positions:		4.7.3.2.24.1		
		1		15 vdc	_____
		2		15 vdc	_____
		3		5 vdc	_____
		4		30 vdc	_____
		5		15 vdc	_____
		6		15 vdc	_____
		7		15 vdc	_____
		8		15 vdc	_____
		9		15 vdc	_____

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	Reference	Limits	Reading
30.2.2.24 (Cont)			
10		15 vdc	
11		15 vdc	
12		15 vdc	
13		15 vdc	
14		15 vdc	
15		15 vdc	
16		15 vdc	
17		15 vdc	
18		15 vdc	
19		40 vdc	
20		40 vdc	
21		40 vdc	
22		30 vdc	
23		30 vdc	
METER FUNCTION switch in position 2	4.7.3.2.24.2		
METER SWITCH positions			
1		30 vdc	
4		30 vdc	
5		15 vdc	
6		15 vdc	
7		15 vdc	
8		15 vdc	
9		15 vdc	
10		15 vdc	
11		15 vdc	
12		15 vdc	
13		15 vdc	
14		15 vdc	
15		15 vdc	

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		Reference	Limits	Reading
30.2.2.24	(Cont)			
	16		15 vdc	_____
	17		15 vdc	_____
	18		15 vdc	_____
	20		40 vdc	_____
	21		40 vdc	_____
	22		15 vdc	_____
	METER FUNCTION switch in position 3	4.7.3.2.24.3		
	METER SWITCH positions			
	5		Approx -25 vdc	_____
	8		Approx -25 vdc	_____
	9		Approx -25 vdc	_____
30.2.2.25	Depth Function Switch	4.7.3.2.25		
	Wafer A, switch at			
	OFF pos.		200,000 ohms min	_____
	20'		5 ohms max	_____
	50'		5 ohms max	_____
	100'		5 ohms max	_____
	Wafer B, switch at			
	OFF pos.		200,000 ohms min	_____
	20'		200,000 ohms min	_____
	50'		5 ohms max	_____
	100'		5 ohms max	_____
	Wafer C, switch at			
	OFF pos.		200,000 ohms min	_____

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		Reference	Limits	Reading
30.2.2.25	(Cont)			
	20'		200,000 ohms min	_____
	50'		20,000 ohms min	_____
	100'		5 ohms max	_____
	Meter function switch continuity	4.7.3.2.25.2		
	Ref. No.			
	1		5 ohms max	_____
	2		5 ohms max	_____
	3		5 ohms max	_____
	4		5 ohms max	_____
	5		Approx. 8,200 ohms	_____
	6		Approx. 1,000 ohms	_____
	7		Approx. 4,700 ohms	_____
	8		Approx. 1,000 ohms	_____
	9		5 ohms max	_____
	10		Approx. 200 ohms	_____
	11		55 to 57 ohms	_____
30.2.2.26	Alternator Regulator	4.7.3.2.26		
	Resistance between J6 pins T and Q	4.7.3.2.26.1	19 to 21 ohms	_____
	Resistance between J6 pin T and red test point	4.7.3.2.26.2	5 ohms max	_____
	Resistance between J6 pin T and black test point	4.7.3.2.26.3	5 ohms max	_____

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		Reference	Limits	Reading
30.2.2.26	(Cont)			
	Panel ammeter indication, METER SWITCH on 20	4.7.3.2.26.5	1.8 to 2.2 amperes	_____
	Panel ammeter indication, METER SWITCH at 21	4.7.3.2.26.6	3.6 to 4.4 amperes	_____
30.2.2.27	Yaw and Pitch Angle Data Diodes	4.7.3.2.27		
	Voltage indication			
	METER SWITCH at 1	4.7.3.2.27.3	1.1 to 1.3 volts dc	_____
	METER SWITCH at 2	4.7.3.2.27.4	9.9 to 10.1 volts dc	_____
	Voltage indication			
	METER SWITCH at 2	4.7.3.2.27.5	1.1 to 1.3 volts dc	_____
	METER SWITCH at 1	4.7.3.2.27.6	9.9 to 10.1 volts dc	_____
	Voltage indication			
	METER SWITCH at 1		-1.1 to -1.3 volts dc	_____
	METER SWITCH at 2		-9.9 to -10.1 volts dc	_____
	Voltage indication			
	METER SWITCH at 2		-1.1 to -1.3 volts dc	_____
	METER SWITCH at 1		-9.9 to -10.1 volts dc	_____

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		Reference	Limits	Reading
30.2.2.28	External Transmitter Load	4.7.3.2.28		
	Resistance between P201 pins:	4.7.3.2.28.1		
	f to h		1.15 to 1.19 ohms	_____
	j to m		1.15 to 1.19 ohms	_____
	R to P		6.44 to 6.70 ohms	_____
	T to S		6.44 to 6.70 ohms	_____
	W to Y		1.91 to 1.99 ohms	_____
	a to c		1.91 to 1.99 ohms	_____
	J to L		142.5 to 157.5 ohms	_____
	H to K		142.5 to 157.5 ohms	_____
	B to D		142.5 to 157.5 ohms	_____
	A to C		142.5 to 157.5 ohms	_____
	n to Z		950 to 1050 ohms	_____
	Resistance between connector pins	4.7.3.2.28.2		
P201 n to J19 (VERT BEAM CHG)			5 ohms max	_____
P201 a to J22 (AMPL NO. 2)			5 ohms max	_____
P201 f to J23 (AMPL NO. 4)			5 ohms max	_____
P201 m to J24 (AMPL NO. 3)			5 ohms max	_____
P201 R to J25 (AMPL NO. 6)			5 ohms max	_____
P201 S to J20 (AMPL NO. 1)			5 ohms max	_____
P201 Y to J21 (AMPL NO. 5)			5 ohms max	_____

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	Reference	Limits	Reading
30.2.2.28 (Cont)			
P201 c to P201 h		5 ohms max	_____
P201 j to P201 h		5 ohms max	_____
P201 P to P201 h		5 ohms max	_____
P201 T to P201 h		5 ohms max	_____
P201 W to P201 h		5 ohms max	_____
P201 d to P201 U		5 ohms max	_____
P201 i to J26 P		5 ohms max	_____

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